

TEST REPORT
UL 1778
STANDARD FOR SAFETY
Uninterruptible Power Systems

Report Number..... : Qst-24N060116-1SR

Date of issue : 2024-06-20

Total number of pages : 85 pages

Name of Testing Laboratory Shenzhen Qstlab Testing Co., Ltd.

preparing the Report..... : No.208,No.4,Lane 1, Tangfang Garden, Xin'an Street, Bao'an
Distic, Shenzhen, Guangdong China

Applicant's name..... : Shandong Goldencell Power Technology Co.,Ltd.

Address : No.X6699 Guangming Boulevard, Hightech District Zaozhuang
City Shandong Province

Test specification:

Standard : UL 1778-2023

Test procedure..... : UL Test Report

Non-standard test method..... : N/A

Test Report Form No..... : UL 1778

Test Report Form(s) Originator.... : UL(US)

Master TRF : Dated 2023

Test item description	UNINTERRUPTIBLE POWER SUPPLY
Trade Mark	GOLDE  MATE
Manufacturer	Shandong Goldencell Power Technology Co.,Ltd. No.X6699 Guangming Boulevard, Hightech District Zaozhuang City Shandong Province
Model/Type reference	UPS 1000VA Pro/800W
Ratings	Lithium-Ion Battery Energy: 230.4Wh(9Ah 25.6V dc) AC input:90-140Vac,1000VA/800W, 50/60Hz, Output:120Vac

Responsible Testing Laboratory (as applicable), testing procedure and testing location(s):		
☒	Testing Laboratory:	Shenzhen Qstlab Testing Co., Ltd.
Testing location/ address		No.208,No.4,Lane 1, Tangfang Garden, Xin'an Street, Bao'an Distric, Shenzhen, Guangdong China
Tested by (name, function, signature)		Karl Yin
Approved by (name, function, signature) ..		Mike Chen
		
Testing procedure: CTF Stage 1:		
Testing location/ address		
Tested by (name, function, signature)		
Approved by (name, function, signature) ..		
Testing procedure: CTF Stage 2:		
Testing location/ address		
Tested by (name + signature)		
Witnessed by (name, function, signature) .		
Approved by (name, function, signature) ..		
Testing procedure: CTF Stage 3:		
Testing procedure: CTF Stage 4:		
Testing location/ address		
Tested by (name, function, signature)		
Witnessed by (name, function, signature) .		
Approved by (name, function, signature) ..		
Supervised by (name, function, signature) :		

List of Attachments (including a total number of pages in each attachment):

Attachment 1: 3 pages (Photo)

Summary of testing:**Tests performed (name of test and test clause):**

The submitted samples were found to comply with the requirements of:

- UL 1778-2023

Testing location:

Shenzhen Qstlab Testing Co., Ltd.

No.208,No.4,Lane 1, Tangfang Garden, Xin'an Street, Bao'an Distric, Shenzhen, Guangdong China

Summary of compliance with National Differences (List of countries addressed):**Statement concerning the uncertainty of the measurement systems used for the tests**

(may be required by the product standard or client)

☐ **Internal procedure used for type testing through which traceability of the measuring uncertainty has been established:****Procedure number, issue date and title:**

Calculations leading to the reported values are on file with the NCB and testing laboratory that conducted the testing.

☒ **Statement not required by the standard used for type testing**

(Note: When IEC or ISO standard requires a statement concerning the uncertainty of the measurement systems used for tests, this should be reported above. The informative text in parenthesis should be delete in both cases after selecting the applicable option)

Copy of marking plate:

The artwork below may be only a draft. The use of certification marks on a product must be authorized by the respective NCBs that own these marks.



Notes:

The above labels are draft of an artwork for marking plate pending approval by National Certification Bodies and it shall not be affixed to products prior to such an approval.

Possible test case verdicts:

- test case does not apply to the test object.....: N/A
- test object does meet the requirement.....: P (Pass)
- test object does not meet the requirement.....: F (Fail)

Testing:

Date of receipt of test item: 2024-06-10

Date (s) of performance of tests: 2024-06-10 to 2024-06-20

General remarks:

"(See Enclosure #)" refers to additional information appended to the report.

"(See appended table)" refers to a table appended to the report.

Throughout this report a ☐ comma / ☒ point is used as the decimal separator.

Manufacturer's Declaration per sub-clause 4.2.5 of IECCE 02:

The application for obtaining a Test Certificate includes more than one factory location and a declaration from the Manufacturer stating that the sample(s) submitted for evaluation is (are) representative of the products from each factory has been provided

☐ Yes

☒ Not applicable

When differences exist; they shall be identified in the General product information section.

Name and address of factory (ies): Shandong Goldencell Power Technology Co.,Ltd.

No.X6699 Guangming Boulevard, Hightech District
Zaozhuang City Shandong Province

General product information and other remarks:

1. The products covered in this report are UNINTERRUPTIBLE POWER SUPPLY
2. All the test on the model UPS 1000VA Pro/800W

UL 1778			
Clause	Requirement + Test	Result - Remark	Verdict

1	General		P
1.1	Scope		P
	Replace this clause of the RD with the following:		P
1.1.1	Equipment covered by this Standard		P
	This Standard applies to UNINTERRUPTIBLE POWER SYSTEMS (UPS). The primary function of the UPS for this Standard is to ensure continuity of an alternating power source. The UNINTERRUPTIBLE POWER SYSTEM may also serve to improve the quality of the power source by keeping it within specified characteristics. This Standard is applicable to movable, stationary, fixed, and built-in UPS for distribution systems up to 600 V a.c. This equipment is designed to be installed in accordance with the Canadian Electrical Code, Part I, CSA C22.1, or the National Electrical Code, ANSI/NFPA 70, and, unless otherwise identified, the Standard for the Protection of Electronic Computer Data-Processing Equipment, ANSI/NFPA 75.		P
	This Standard specifies requirements intended to ensure safety for the OPERATOR and, where specifically stated, for SERVICE PERSONNEL.		P
	This Standard is intended to reduce the risk of fire, electric shock, or injury to persons from installed equipment, both as a single unit or as a system of interconnected units, subject to installing, operating, and maintaining the equipment in the manner prescribed by the manufacturer.		P
1.1.2	Additional requirements		P
	In addition to the requirements in this Standard, a UPS is to comply with the UPS-relevant requirements of CAN/CSA-C22.2 No. 60950-1/UL 60950-1, March 2007, second edition, Information Technology Equipment – Safety – Part 1: General requirements (RD), as applicable for the country where the product will be used. Wherever there is a conflict between the requirements of this Standard and the RD, the requirements of this Standard will prevail.		P
	Engine-driven d.c. power generators intended to provide backup power for the battery supply circuit of UPS units are investigated for compliance with the requirements of UL 2200, and CSA C22.2 No. 100.		P
	UPS that employ hospital grade components identified by the markings "Hospital Only", "Hospital Grade", or a green dot on the BODY of the component, or otherwise implying suitability for medical use, are evaluated to the requirements of this Standard and CAN/CSA-C22.2 No. 60601-1/UL 60601-1.		P

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	Requirements additional to those specified in this Standard may be necessary for equipment intended for use where ingress of water is possible; for guidance on such requirements and on relevant testing, see Annex LLL and Annex T/RD.		P
1.1.3	Exclusions		P
	These requirements do not cover UPS units for use as legally required standby systems, described in Article 701 of the National Electrical Code, ANSI/NFPA 70, and emergency power supply described in Section 46 of the Canadian Electrical Code, Part I, CSA C22.1. See Annex LLL.		P
	Where considered appropriate, revision of requirements will be proposed and adopted in conformance with the methods employed for development, revision, and implementation of this Standard.		P
1.1.4	Additional applications		P
	This Standard does not cover all types of UPS, but it may be taken as a guide for such equipment. Requirements additional to those specified in this Standard are in some cases necessary for specific applications, e.g.,		P
	a) equipment intended for operation while exposed to conditions such as extremes of temperature; excessive dust, moisture, or vibration; flammable gases; or corrosive or explosive atmospheres;		P
	b) UPS equipment based on rotary machinery;		P
	c) UPS equipment meeting emergency lighting and power requirements as specified in UL 924; and central power system described in CSA C22.2 No 141-10.		P
1.1.101	Normative references		P
	Any undated reference to a code or standard appearing in the requirements of this Standard shall be interpreted as referring to the latest edition of that code or standard.		P
	Products covered by this Standard are to comply with the reference installation codes and standards as noted (in Annex III) as appropriate for the country where the product is to be used. When the product is intended for use in more than one country, the product is to comply with the installation codes and standards for all countries where it is intended to be used.		P
	For products intended for use in Canada, general requirements are given in CAN/CSA-C22.2 No. 0.		P
1.2	Definitions		P
	The provisions of 1.2/RD apply, together with the following:		P

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	For the purpose of this Standard, the following definitions apply. Where the terms "voltage" and "current" are used, they imply the rms values, unless otherwise specified.		P
1.2.1	Equipment electrical ratings		P
1.2.1.101	UNINTERRUPTIBLE POWER SYSTEM (UPS) : Combination of converters, switches, and energy storage devices (such as batteries), constituting a power system for maintaining continuity of power to load in case of input POWER FAILURE.		P
1.2.1.102	BYPASS : Power path alternative, either internal or external, to the UPS.		P
1.2.1.103	POWER FAILURE : Any variation in power supply that can cause unacceptable performance of the load equipment.		P
1.2.1.104	PRIMARY POWER : Power that is normally available, as supplied by an electrical utility company or by a USER'S generator		P
1.2.1.105	APPARENT OUTPUT POWER : The product of the rms output voltage and rms current. It is given for a load in VA or kVA, with a specified POWER FACTOR.		P
1.2.1.106	ACTIVE POWER : Sum of the electrical power at the fundamental frequency and the powers of each harmonic component from the output terminals, in W or kW.		P
1.2.1.107	POWER FACTOR: Characteristic of an a.c. load expressed as the ratio of ACTIVE POWER to apparent power.		P
1.2.1.108	RATED APPARENT OUTPUT POWER : Apparent output power as declared by the manufacturer		P
1.2.1.109	RATED ACTIVE OUTPUT POWER : Active output power as declared by the manufacturer.		P
1.2.1.110	BACKFEED : The condition in which HAZARDOUS VOLTAGE or energy available within the UPS is fed back to any of the input terminals, either directly or by a leakage path, when operating in the STORED ENERGY MODE with PRIMARY POWER disconnected.		P
1.2.1.111	BACKFEED PROTECTION : The method used to reduce the risk of electric shock due to BACKFEED		P
1.2.1.112	ELECTRONIC BACKFEED PROTECTION : BACKFEED PROTECTION that does not employ an air gap.		P
1.2.1.113	FIELD-INSTALLABLE ACCESSORY EQUIPMENT : Equipment intended to modify the UPS construction or function. Such equipment is typically installed in		P

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Clause	Requirement + Test	Result - Remark	Verdict
	the field by service personnel.		
1.2.2	Operating conditions		P
	Replace this definition of the RD with the following:		P
1.2.2.1	NORMAL LOAD : The mode of operation that approximates as closely as possible the rated conditions of normal use in accordance with the manufacturer's operating instructions.		P
	Delete this definition in the RD:		P
1.2.2.3	RATED RESTING TIME		P
	This definition is not applicable. Any reference to RATED RESTING TIME in the RD is not applicable.		P
	Additional definitions:		P
1.2.2.101	LINEAR LOAD : A load in which the current drawn from the supply is defined by the following relationship: $I = U/Z$ where: I = load current U = supply voltage Z = load impedance		P
1.2.2.102	NON-LINEAR LOAD : A load in which the parameter Z (load impedance) is no longer a constant but is a variable dependent on other parameters, such as voltage or time (see Annex BBB).		P
	STORED ENERGY MODE : The operation of the UPS when supplied by the following conditions:		P
	a) PRIMARY POWER is disconnected or is out of a given tolerance;		P
	b) battery stored energy source being discharged;		P
	c) load is within the given range; and		P
	d) output voltage is within the given tolerance.		P
1.2.2.104	BATTERY, VALVE-REGULATED : A battery in which the venting of the products of electrolysis is controlled by a reclosing pressure-sensitive valve.		P
1.2.2.105	BATTERY, VENTED : A battery in which the products of electrolysis and evaporation are allowed to escape freely to the atmosphere. Also known as "flooded" or "wet".		P
1.2.8	Circuits and circuit characteristics		P
	Additional definitions:		P
1.2.8.101	BRANCH CIRCUIT : The portion of the building wiring system beyond the final overcurrent protective device on the power-distribution panel that protects the circuit to the field-wiring terminals of PERMANENTLY CONNECTED EQUIPMENT or the receptacle outlet of PLUGGABLE EQUIPMENT TYPE A or PLUGGABLE EQUIPMENT TYPE B.		P
1.2.13	Miscellaneous		P
	Additional definitions:		P

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1.2.13.101	CONTROLLED ENVIRONMENT : An environment that is an indoor, temperature-regulated location such as a computer room, office, or factory floor that is relatively free of conductive contaminants such as carbon dust and the like.		P
1.2.13.102	SHORT-CIRCUIT WITHSTAND RATING : The highest current at rated voltage that a device is intended to interrupt under standard test conditions.		P
1.2.13.103	BRACE : A mechanical assembly that is secured to bus bars to restrict relative motion between the bus bars.		P
1.2.13.104	SUPPORT : A mechanical assembly that is secured to bus bars and that is further secured to a rigid structural member such as the enclosure or a separate member secured to the framework.		P
1.3	General requirements		P
	The provisions of 1.3/RD apply, together with the following:		P
	Additional subclause:		P
1.3.101	The frame or chassis of the equipment shall not be used to carry current other than the allowable touch current as determined in the test specified in 5.101/RD.		P
1.4	General conditions for tests		P
	The provisions of 1.4.1/RD, 1.4.2/RD, 1.4.3/RD, 1.4.4/RD, 1.4.5/RD, 1.4.6/RD, 1.4.7/RD, 1.4.8/RD, 1.4.10/RD, 1.4.11/RD, 1.4.12/RD, 1.4.13/RD, 1.4.14/RD and 1.4.15/RD apply, together with the following:		P
1.4.1	Application of tests		P
	Addition:		P
	Only the leakage current (TOUCH CURRENT) and heating tests shall be performed at input tolerances. Except as noted in other clauses of the Standard, tests shall be conducted at nominal input voltages.		P
1.4.10	Loading configuration of the EUT		P
	Addition to first paragraph:		P
	– loads due to recharging batteries;		P
1.4.15.101	Routine tests		P
	Each production unit shall be subjected to the test requirements specified in Annex PPP.		P
	Components		P
	The provisions of 1.5.1/RD, 1.5.2/RD, 1.5.3/RD, 1.5.4/RD, 1.5.5/RD, 1.5.6/RD, 1.5.7/RD, 1.5.7.1/RD, 1.5.7.2/RD, 1.5.7.3/RD, and 1.5.8/RD apply, together with the following:		P

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Clause	Requirement + Test	Result - Remark	Verdict
	General		P
	Add the following text to the last paragraph of this subclause:		P
	Except as indicated in 1.5, a component of a product covered by this Standard shall comply with the requirements for that component. See Annex III for a list of Standards covering components generally used in the products covered by this Standard. A component shall comply with CSA Group or Underwriters Laboratories Inc. Standards, as appropriate for the country where the product is to be used.		P
	Additional subclauses:		P
	Oil-filled and inverter storage capacitors shall comply with the requirements of Annex AAA. Batteries shall comply with the requirements of 4.3.8.101. Power switching devices shall comply with the requirements of Annex AAA. Relays shall comply with the requirements of Annex AAA. Static transfer devices shall comply with the requirements of Annex AAA. BYPASS/maintenance BYPASS devices shall comply with the requirements of Annex AAA.		N/A
	Capacitors or EMI filters connected across the AC MAINS SUPPLY shall comply with the requirements for across-the-line capacitors in UL 60384-14 or CSA E60384-14, UL 810 or CSA C22.2 No. 190, and UL 1283 or CSA C22.2 No. 8, or the test specified in 5.2.101 of this Standard.		N/A
	FIELD-INSTALLABLE ACCESSORY EQUIPMENT		N/A
	FIELD-INSTALLABLE ACCESSORY EQUIPMENT shall be constructed so that it does not present any hazard specified in this Standard. FIELD-INSTALLABLE ACCESSORY EQUIPMENT shall be installed per manufacturer instructions and, once installed, shall not present any hazard specified in this Standard.		N/A
	The installation of FIELD-INSTALLABLE ACCESSORY EQUIPMENT by an operator shall be restricted to an arrangement that can be accomplished mechanically a tool not provided by the manufacturer and electrically by means of plug-in connections to receptacles available on the basic unit or as a part of the building wiring.		N/A
	The installation of FIELD-INSTALLABLE ACCESSORY EQUIPMENT by qualified service personnel shall be such that:		N/A
	a) The mechanical positioning can be accomplished by means of regular tools normally available at installation or by means of special tools provided as a part of the installation kit by the organization responsible for the product; and		N/A
	b) The electrical connections can be readily accomplished by making use of existing terminals		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	and connections in the unit wherever possible.		
	The requirement in 1.5.108.3 does not preclude the addition or removal of components or insulated conductors or rerouting of insulated conductors to accomplish the desired change as long as the alterations in the unit wiring:		N/A
	a)Can be accomplished by the use of materials and reference to instructions, both of which are furnished as part of the accessory-equipment kit and conversion-unit kit; and		N/A
	b)Do not require the use of makeshift or substitute parts not used in the basic construction of the system.		N/A
	All wiring provided as a part of an item of FIELD-INSTALLABLE ACCESSORY EQUIPMENT or related to its installation shall be acceptable for use at the highest voltage and temperature that can be encountered in the area in which the wire is to be installed.		N/A
	FIELD-INSTALLABLE ACCESSORY EQUIPMENT shall meet the applicable construction and performance requirements of this Standard.		P
	FIELD-INSTALLABLE ACCESSORY EQUIPMENT marking		P
	Each piece of FIELD-INSTALLABLE ACCESSORY EQUIPMENT shall be marked with the manufacturer's name, trademark, or other descriptive marking by means of which the organization responsible for the product can readily be identified and with a distinctive catalogue number or equivalent identification. Alternatively, the marking for a piece of FIELD-INSTALLABLE ACCESSORY EQUIPMENT may be on the package.		P
	FIELD-INSTALLABLE ACCESSORY EQUIPMENT intended to be installed by an operator shall be marked		P
	to indicate the unit for which it is intended or shall be marked with a reference to an instruction manual that tabulates the units for which it is intended. Additionally, FIELD-INSTALLABLE ACCESSORY EQUIPMENT shall include instructions on or packed with the equipment such that the FIELD-INSTALLABLE ACCESSORY EQUIPMENT can be properly mounted on and interconnected with the basic unit.		P
	FIELD-INSTALLABLE ACCESSORY EQUIPMENT intended to be installed by service personnel and FIELD-INSTALLABLE ACCESSORY EQUIPMENT shall include instructions either on or packed with each piece of FIELD-INSTALLABLE ACCESSORY EQUIPMENT. The instructions shall provide a detailed sequence of the mechanical and electrical steps that are necessary for proper installation and operation.		P
	Power interface		P

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Clause	Requirement + Test	Result - Remark	Verdict
	<i>The provisions of 1.6.1/RD, 1.6.2/RD, and 1.6.4/RD apply, together with the following:</i>		P
	<i>Additional subclause:</i>		P
	Telecommunication UPS intended for either permanent or cord connection to single or polyphase 600 V nominal or less ac supply		P
	All pole-mounted CATV UPS enclosures shall have provision for connecting three 6 AWG copper bonding or grounding conductors as follows:		P
	a)bonding to the cable suspension strand or support wire on the pole;		P
	b)bonding to the ac input power-system grounded neutral; and		P
	c)grounding to the ground rod.		P
	All pedestal-mounted CATV UPS enclosures shall have provision for connecting two 6 AWG copper bonding or grounding conductors as follows:		P
	a)bonding to the ac input power-system grounded neutral; and		P
	b)grounding to the ground rod.		P
1.6.2	Input current		P
	<i>Additional subclauses:</i>		P
	While supplying rated output under each of the conditions described in Items (a) to (d), the input current shall not be more than 110 % of the rated value:		P
	a)Recharging mode: The UPS shall receive power from the PRIMARY POWER source while delivering maximum rated alternating current power and the battery charging current.		P
	b)Stored energy mode: For a UPS used with a remote battery, while simulating PRIMARY POWER outage, the inverter portion of the UPS shall receive power from either a fully charged battery bank or an external dc source of supply and shall be allowed to deliver maximum rated alternating current power. The input dc current shall be measured.		P
	c)BYPASS mode: The transfer switch shall be positioned to allow the PRIMARY POWER for the output load to bypass the rectifier/charger and inverter sections of the UPS and be delivered directly to the load adjusted to draw maximum rated alternating current power.		N/A
	d)Normal mode: With a fully charged battery, the UPS shall receive power from the PRIMARY POWER source and deliver maximum rated alternating current power. With reference to 1.6.2.101 (a), the battery charging circuit shall be connected to one of the following:		N/A
	a)a resistive-capacitive (RC) load having capacitance of 1000 microfarads per output dc ampere rating of the battery supply;		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	b)a battery supplemented with a resistive load bank; or		N/A
	c)a battery having an ampere-hour (watt or kilowatt) and voltage rating corresponding to that which is intended to be used with the UPS.		N/A
	If a battery load is used as described in 1.6.2.102 (c), the battery shall be prepared for charging by first connecting it to the proper load and then discharging it to the low-voltage disconnect (LVD) potential, or 80 % of the float charging voltage rating of the battery for a UPS having a low-voltage disconnect lockout, at a rate not exceeding the discharge rate assigned by the battery manufacturer.		N/A
	Markings and instructions		P
	<i>The provisions of 1.7.1/RD, 1.7.2/RD, 1.7.4/RD, 1.7.5/RD, 1.7.6/RD, 1.7.7/RD, 1.7.8/RD, 1.7.9/RD, 1.7.10/RD, 1.7.11/RD, 1.7.12/RD, 1.7.13/RD, and 1.7.14/RD apply, together with the following:</i>		P
	<i>Additional Note:</i> NOTE 101 In Canada, there are two official languages: English and French. Annex JJJ provides examples of French translations of the on-product markings specified in this Standard. Markings required by this Standard may have to be provided in other languages to conform with the language requirements of the country where the product is to be used.		P
	Power rating		P
	<i>Additional paragraph:</i>		P
	The marking shall include the following:		P
	a)number of phases, unless intended for single phase only;		P
	b)RATED ACTIVE OUTPUT POWER, in W or kW;		P
	c)RATED APPARENT OUTPUT POWER, in VA or kVA;		P
	d)rated output voltage;		P
	e)rated output current;		P
	f)rated output frequency;		P
	g)where applicable, short-circuit withstand rating (see 5.101); and		P
	h)if appropriate, the unbalanced load capability.		P
	New subclauses:		P
	One of the following markings shall be placed on units that exceed the maximum output voltage harmonics when measured in accordance with Annex DDD:		P
	a)NOTICE: The output of this device is not purely sinusoidal. It has a nominal total voltage harmonic distortion of percent, with the nominal value of the largest single voltage harmonic of percent.		P
	b)NOTICE: For use with loads.		P

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	c)NOTICE: Only for use with manufacturer: ; model:		P
	For the marking in item (b), the generic type of load shall be inserted in the blank space provided. Examples of such generic load types include computer loads and information processing loads.		P
	For units designed with additional separate automatic BYPASS/maintenance BYPASS, additional input a.c. supply, or external batteries, relevant supply ratings shall be allowed to be specified in the accompanying installation instructions. Where this is done, the following instruction shall appear on or near the point of connection:		P
	SEE INSTALLATION INSTRUCTIONS BEFORE CONNECTING TO THE SUPPLY		P
	Compliance is determined by inspection.		P
	Industrial-type output receptacles with standard configurations according to the CEC, Part I and the NEC may have output voltage, frequency, or types of current different from the rating of the receptacles, provided that a highly visible marking indicating the voltage, frequency, and current is permanently applied adjacent to each output receptacle or group of output receptacles, such that no hazard will result from a misunderstanding of the intended output rating. See Annex LLL.		P
	The date of manufacture shall be plainly and permanently marked on a unit such that the date is readily visible after installation of the unit. This date may be abbreviated, or in a code affirmed by the manufacturer, provided that the code does not		P
	a)repeat in less than 20 years; and		P
	b)require reference to the production records of the manufacturer to determine when the unit was manufactured.		P
	In Canada, when a CATV UPS is used by public utilities, the following marking or equivalent shall appear on the UPS:		P
	"This UPS is for use with telephone equipment in accordance with section 60 of the <i>Canadian Electrical Code, Part I</i> , and is subject to inspection by an inspector".		P
	The following marking shall appear on all CATV UPS:		P
	"This unit is intended for connection to pole-mounted or underground amplifiers".		P
	CATV UPS that are not provided with a separate enclosed service switch or circuit breaker complying with CSA C22.2 No. 4 or CSA C22.2 No. 5 shall be marked with the following or equivalent:		P
	"In order to comply with the <i>Canadian Electrical Code, Part I</i> , disconnecting means for the CATV UPS shall be marked as being suitable for use as service equipment".		P
	Terminals intended for connection of batteries shall		P

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Clause	Requirement + Test	Result - Remark	Verdict
	indicate the polarity according to IEC 60417, or be so constructed as to prevent improper connection.		
	Safety instructions and marking		P
	<i>Additional subclauses:</i>		P
	Ambient temperature		P
	The maximum ambient operating temperature shall be indicated in the instruction manual.		P
	Signalling circuits		P
	Information shall be provided in the installation instructions as to the purpose and connection of any signalling circuits, relay contacts, emergency power off (EPO) circuits, etc. Attention should be drawn to the necessity of maintaining the security of any SELV CIRCUIT when connected to other equipment.		P
	Internal circuit configuration		P
	Installation instructions shall carry sufficient information, including the basic internal circuit configuration of the UPS, to emphasize its compatibility with power distribution systems (see Annex V/RD).		P
	Special attention shall be given to compatibility with the relevant wiring rules and to BYPASS circuits.		P
	General instructions		P
	The information in items (a) – (s), as appropriate, shall be provided for a UPS, a remote battery supply/cabinet assembly, and a maintenance BYPASS cabinet assembly. A single installation manual may be used for a UPS investigated in combination with a remote battery supply/cabinet or maintenance BYPASS cabinet assembly. The information in items (c) – (s) may be marked on the unit in lieu of providing it in the instruction manual.		P
	IMPORTANT SAFETY INSTRUCTIONS		P
	a)SAVE THESE INSTRUCTIONS – This manual contains important instructions for Models _____ that should be followed during installation and maintenance of the UPS and batteries. (<i>Blank space is to be filled in with appropriate model numbers.</i>)		P
	b)If pressure terminal connectors or the fastening hardware is not provided on the UPS as shipped, the instruction manual shall indicate which pressure terminal connector or component terminal assemblies are for use with the UPS.		P
	c)With reference to item (b), the terminal assembly packages and the instruction manual shall include information identifying wire size and manufacturer's name, trademark, or other descriptive marking by which the organization responsible for the product can be identified.		P

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Clause	Requirement + Test	Result - Remark	Verdict
	d)If a pressure terminal connector provided in the unit or in a terminal assembly for a field installed conductor requires the use of other than an ordinary tool for securing the conductor, identification of the tool and any necessary instructions for using the tool shall be included in the instruction manual.		P
	e)A unit provided with a wire connector for field installed wiring shall be provided with instructions specifying that the connector provided is to be used in making the field connection.		P
	f)A unit employing pressure terminal connectors for field wiring connections shall be provided with instructions specifying a range of values or a nominal value of tightening torque to be applied to the clamping screws of the terminal connectors.		P
	g)The instruction manual for stationary and portable units having multiple input voltage ratings shall include information indicating the type of attachment plug that is to be used for connection to a voltage supply other than what the UPS is set for when it is shipped from the factory.		P
	h)The instruction manual for a UPS that exceeds the temperature limits of Table 4C/RD shall specify that the unit is to be installed so that it is not likely to be contacted by people.		P
	i)The instruction manual for a UPS shall contain complete instructions concerning proper selection of the power supply cord when the UPS is intended for use with an appliance coupler and is not provided with a detachable power cord.		P
	j)Instructions for field assembly of modules of a modular unit, including an interconnection wiring diagram, shall be either		P
	1)packaged with the modules; or		P
	2)contained in the instruction manual, provided that the marking on the module makes reference to the instruction manual.		P
	k)Remote dc sources, including batteries, not provided with overcurrent protection, shall include a statement in the installation instruction manual indicating that overcurrent protection is to be provided by others. See 2.7.101.		P
	l)UPS with permanently connected ac outputs, not provided with overcurrent protection, shall include a statement in the installation instruction manual indicating that overcurrent protection is to be provided by others.		P
	m)The instruction manual shall include information identifying the number of conductors and range of conductor sizes for a unit having a single equipment field-wiring terminal that is intended for connection of more than one conductor.		P
	n)For a unit provided with field-wiring terminals or leads, the instruction manual shall include the information indicated in Row 1, 2, 3, or 4 of Table 1.7.2.104 , or with equivalent wording, if it is		P

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Clause	Requirement + Test	Result - Remark	Verdict
	1)intended for use on a supply circuit rated 110 A or less; or		P
	2)intended for field connection with 1 AWG (42.4 mm ²) or smaller conductors.		P
1.7.2.104	Termination markings		N/A
	o)For a unit provided with field-wiring terminals or leads, the instruction manual shall include the information indicated in Row 3 or 4 of Table 1.7.2.104 , or with equivalent wording, if it is		N/A
	1)intended for use on a supply circuit rated more than 110 A; or		N/A
	2)intended for field connection with conductors larger than 1 AWG (42.4 mm ²).		N/A
	p)If applicable, the instruction manual shall include a statement indicating that Class 1 wiring methods are to be used for field wiring connections to terminals of a Class 2 circuit.		N/A
	q)the instruction manual for a unit investigated for use in a CONTROLLED ENVIRONMENT shall indicate that the unit is intended for installation in a temperature-regulated, indoor area that is relatively free of conductive contaminants.		N/A
	r)If an abnormal test is terminated by operation of the intended BRANCH CIRCUIT overcurrent protective device; or if the a.c. input overcurrent protection is relied upon for protection of an a.c. output receptacle, the instruction manual for a unit thus described shall include the word "CAUTION" and the following or equivalent: "To reduce the risk of fire, connect only to a circuit provided with ____A maximum branch circuit overcurrent protection in accordance with the <i>National Electrical Code</i> , <i>ANSI/NFPA 70</i> and the <i>Canadian Electrical Code, Part I, C22.1</i> ". (The blank space is to be filled in with the appropriate ampere rating of BRANCH CIRCUIT overcurrent protection).		N/A
	s)Instructions shall be provided that explain how to install a remote shunt-trip circuit breaker for a unit that is intended to be used with this type of breaker.		N/A
	t)The instruction manual for a UPS intended to be used with a remote battery supply that is not provided with the UPS shall make reference to the battery manufacturer's installation manual for battery installation and maintenance instructions.		N/A
	Installation instructions for CATV UPS		N/A
	The instruction manual for a UPS intended to supply power to cable TV equipment shall include the instructions for proper installation. The text of these instructions shall appear under the heading "Installation Instructions".		N/A
	Installation instructions for modular units		N/A
	Individual modules of a modular unit may be of the open construction type – either no enclosure or a partial enclosure is supplied – provided that when the modules are assembled together in the field as		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	intended, the unit enclosure complies with the requirements in this Standard and in the RD. Identification of the modules and instructions for assembling shall be provided.		
1.7.8	Controls and indicators		P
1.7.8.3	Symbols		P
	<i>Additional paragraph:</i>		P
	<i>Safety symbols and their meanings are shown in Annex CCC.</i>		P
1.7.13	Replaceable batteries		N/A
	<i>Additional subclauses:</i>		N/A
	Marking requirements for batteries located in a SERVICE ACCESS AREA		N/A
	Battery cabinets or compartments shall be provided with the following clearly legible information, fixed upon the battery cabinet or compartment of stationary batteries, in such a position that it can be clearly seen by SERVICE PERSONNEL before accessing the battery compartment:		N/A
	a) battery manufacturer name, catalogue number, and number of batteries;		N/A
	b) nominal voltage of total battery string;		N/A
	c) nominal capacity of total battery string;		N/A
	d) a caution label denoting an energy and chemical hazard and reference to the maintenance handling and disposal instructions for the safety of SERVICE PERSONNEL;		N/A
	e) for a UPS having a battery supply without transformer isolation, a label containing the word "CAUTION" and the following warning or equivalent: "Risk of Electric Shock – Battery Circuit is not isolated from ac input, hazardous voltage may exist between battery terminals and ground. Test before touching."		N/A
	Instructions		P
	Internally mounted battery		P
	Instructions shall carry sufficient information to enable the replacement of the battery with a suitable manufacturer and catalogue number.		P
	Safety instructions to allow access by SERVICE PERSONNEL shall be stated in the installation/service handbook.		P
	If batteries are to be installed by SERVICE PERSONNEL, instructions for interconnections, including terminal torque, shall be provided.		P
	Externally mounted battery		N/A
	Installation instructions shall state voltage, ampere-hour (watt or kilowatt) rating, charging information, method of protection, and maximum fault current required on installation to coordinate with UPS protective devices.		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Where the battery is not provided by the UPS manufacturer, the installation and maintenance instructions shall be provided by the battery manufacturer.		N/A
	External battery cabinets		N/A
	External battery cabinets supplied with the UPS shall have adequate installation instructions to define cable sizes for connection to the UPS. Installation instructions for the battery shall be provided by the UPS battery manufacturer.		N/A
	Installation or replacement of batteries located in an OPERATOR ACCESS AREA		N/A
	The batteries shall be of a sealed or valve-regulated type.		N/A
	Reverse connection of the terminals shall be prevented if reverse connection could result in a hazard within the meaning of this Standard.		N/A
	The OPERATOR manual shall include the following instructions for battery replacement and disposal:		N/A
	When replacing batteries, replace with the same number of the following battery: __ (The blank space shall be filled with the manufacturer's name and the model number of the batteries used with the equipment.)		N/A
	Installation or replacement of batteries located in a SERVICE ACCESS AREA		N/A
	Bare parts that involve an energy hazard shall be located, enclosed, guarded, or provided with a barrier to take into account the unintentional bridging by conductive materials that might be present during service operations.		N/A
	Bare parts operating at a HAZARDOUS VOLTAGE level shall be located or guarded such that unintentional contact with such parts is unlikely during servicing operations involving other parts of the equipment.		N/A
	The service manual shall include the following instructions for battery replacement and disposal:		N/A
	Servicing of batteries should be performed or supervised by personnel knowledgeable about batteries and the required precautions.		N/A
	When replacing batteries, replace with the same type and number of batteries or battery packs.		N/A
	a) Remove watches, rings, or other metal objects		N/A
	b) Use tools with insulated handles.		N/A
	c) Wear rubber gloves and boots.		N/A
	d) Do not lay tools or metal parts on top of batteries.		N/A
	e) Disconnect charging source and load prior to installing or maintaining the battery.		N/A
	f) Disconnect charging source and load prior to installing or maintaining the battery.		N/A
	VENTED BATTERIES		N/A
	UPS having VENTED BATTERIES shall have an additional caution in the manual describing the safe		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	methods of handling VENTED BATTERIES.		
2	Protection from hazards		P
	The provisions of 2/RD apply, together with the following.		P
	Protection from electric shock and energy hazards		P
	<i>The provisions of 2.1.1/RD, 2.1.1.1/RD, 2.1.1.2/RD, 2.1.1.3/RD, 2.1.1.4/RD, 2.1.1.5/RD, 2.1.1.6/RD, 2.1.1.7/RD, 2.1.1.8/RD, 2.1.2/RD, and 2.1.3/RD apply, together with the following additions.</i>		P
	<i>Additional subclauses:</i>		P
	BACKFEED PROTECTION		P
	HAZARDOUS VOLTAGE and/or energy shall not appear on the a.c. input terminals under the condition of a single fault on a component (such as in the control circuit) upon loss of the a.c. input voltage.		P
	For fixed-installation UPS, BACKFEED PROTECTION may be provided internally or externally to the UPS in the a.c. input line.		N/A
	When a BACKFEED PROTECTION isolator is external to the UPS, the supplier shall specify in the installation instructions the suitable isolating device to be used.		P
	The UPS shall be tested as specified in 2.1.102 . No shock hazard shall exist at the a.c. input when measured 1 s after de-energization of the a.c. input line if the equipment is PLUGGABLE TYPE A, or 15 s if permanently connected or PLUGGABLE TYPE B.		P
	The UPS shall be operated with an a.c. power output current at no load and full load and with a single fault applied to the component being investigated. The fault placed on the component shall simulate the failure mode of that component. The a.c. input line shall then be disconnected and any BACKFEED measured.		P
	<i>Compliance is checked by inspection of the equipment and relevant circuit diagram and by simulating fault conditions according to 5.3.7/RD and other sections of this Standard, such as Annex FFE.</i>		P
	In the United States the following applies:		P
	A UPS provided with a marking indicating that it is suitable for use in a computer room shall comply with Annex NAE 3.4.11/RD.		P
	Provisions for earthing and bonding		P
2.6.3	Protective earthing conductors and protective bonding conductors		P
	Size of protective bonding conductors		P
	Resistance of earthing conductors and their terminations		P
	<i>Replace item b) of the RD with the following: b) For equipment powered an AC MAINS SUPPLY, if the</i>		P

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Clause	Requirement + Test	Result - Remark	Verdict
	<i>PROTECTIVE CURRENT RATING of the circuit under test exceeds 16 A, the test current is 200% of the PROTECTIVE CURRENT RATING, up to a maximum test current of 500 A, and the duration of the test is as shown in Table 2E.</i>		
2.6.101	Bonding		P
	For Class I PLUGGABLE EQUIPMENT TYPE A, the UPS shall provide sufficient terminals, earthed socket- outlets, or other means to permit, in the final installed system configuration, equipotential protective bonding to the UPS from other Class I equipment, including external UPS battery cabinets, irrespective of whether the UPS primary protective conductor is disconnected from its source. Any special bonding instructions shall be stated in the user's instructions.		P
	Refer to Annex EEE for grounding supplied from separately derived a.c. sources.		P
	A UPS having a separately derived output that is intended to be grounded in the field shall be marked with the following or equivalent words:		P
	"The output a.c. circuit is considered as a separately derived source".		P
	If local codes require grounding of this circuit, use terminal (identify terminal) for bonding this circuit to the ENCLOSURE. Ground the ENCLOSURE to a suitable grounding electrode in accordance with local code requirements.		P
	Overcurrent and earth fault protection in primary circuits		P
	Basic requirements		P
	<i>Additional paragraphs:</i>		P
	For components in series with the AC MAINS SUPPLY to the equipment, such as the supply cord, appliance coupler, EMI filter, and switch, short-circuit and earth fault protection may be provided by protective devices in the building installation. See Annex LLL.		P
	AC output overcurrent protection shall be provided. Overcurrent protection need not be provided for permanently connected equipment when the overcurrent protection is provided by others in the final installation.		P
	If reliance is placed on protection in the building installation, the installation instructions shall comply with 2.7.1/RD, except that for PLUGGABLE EQUIPMENT TYPE A, the buildings installation shall be regarded as providing protection in accordance with the rating of the wall outlet, and 2.7.1/RD does not apply.		P
	If the protection against electric shock of the UPS (see 2.1) relies on residual current devices in the building installation circuit and the design of the UPS is such that in any normal or abnormal		P

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Clause	Requirement + Test	Result - Remark	Verdict
	operating condition a fault current to earth with d.c. component is possible, the installation instructions shall define (an) appropriate building residual current device(s).		
	<i>Compliance is checked by inspection and functional test.</i>		P
	Battery supply circuit overcurrent protection		P
	A battery supply circuit shall be provided with overcurrent and short-circuit protection as follows:		P
	Where the batteries are installed inside the UPS, the battery supply circuit shall be provided with a protective device located adjacent to the battery connecting means before any component that may fail short-circuited, such as capacitors, solid-state devices, or similar components.		P
	Where the batteries are installed outside the UPS, the location of the overcurrent-protective device shall be as indicated above, within the separate cabinet.		P
	The rating of the overcurrent-protective device located internally shall be such as to protect against conditions described in 5.3/RD.		P
	For a UPS to be used with a separated battery supply, the rating of the overcurrent-protective device shall be indicated in the instruction manual.		P
	When the overcurrent-protective device is not provided by the manufacturer, suitable installation instructions shall be provided for installation of an overcurrent-protective device.		P
	<i>Compliance is checked by inspection and tests.</i>		P
	Rating of protective device		P
	The rating of the overcurrent protective device located internally shall be such as to protect against conditions described in 5.3.1/RD.		P
	<i>Compliance is checked by inspection and tests</i>		P
	Type and location of overcurrent protective devices		P
	Overcurrent protection located within the UPS or stationary remote battery cabinet may be a supplementary overcurrent protective device or branch circuit protective device. Overcurrent devices in permanently connected battery cabinets shall be branch circuit protective devices.		P
	Safety interlocks		P
	<i>Additional subclause:</i>		P
	Protection of service personnel		P
	General		P
	In addition to the requirements of 2.8/RD, clauses 2.8.101.2 to 2.8.101.8 apply to SERVICE PERSONS who find it necessary to reach over, under, across, and around an uninsulated electrical part or moving part to make adjustments or measurements while the UPS is energized.		P

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Clause	Requirement + Test	Result - Remark	Verdict
	Covers		P
	Parts at HAZARDOUS VOLTAGE or energy level shall be so arranged and covers so located as to reduce the risk of electric shock or high current levels while covers are being removed and replaced.		P
	Location and guarding of parts		P
	Parts at HAZARDOUS VOLTAGE or energy level and moving parts that involve a risk of injury to persons shall be located, guarded, or enclosed so as to reduce the likelihood of unintentional contact by a SERVICE PERSON adjusting or resetting controls, or the like, or performing mechanical functions that may be performed with the UPS energized, such as lubricating a motor, adjusting the setting of a control with or without marked dial settings, resetting a trip mechanism, or operating a manual switch.		P
	Parts on doors		P
	Parts at HAZARDOUS VOLTAGE or energy level, located on the rear side of a door, shall be guarded or insulated to reduce the likelihood of unintentional contact of the live parts by a SERVICE PERSON.		P
	Compliance with 2.8.101.2 to 2.8.101.4 is checked by inspection, measurement, and use of the test finger (Figure 2A/RD).		P
	Component access		P
	A component that requires inspection, resetting, adjustment, servicing, or maintenance while energized shall be so located and mounted with respect to other components and with respect to grounded metal parts that it is accessible for electrical service functions without subjecting the SERVICE PERSON to the risk of electric shock, HAZARDOUS ENERGY LEVEL, high current, or injury to person by adjacent moving parts. Access to a component shall not be impeded by other components or wiring.		P
	For an adjustment that is to be made with a screwdriver or similar TOOL when the UPS is energized, the requirement in 2.8.3/RD necessitates that protection be provided so that inadvertent contact with adjacent uninsulated hazardous live parts involving a risk of electric shock or HAZARDOUS ENERGY LEVEL is unlikely, taking into consideration that misalignment of the TOOL with the adjustment can result when an adjustment is attempted.		P
	This protection shall be provided by		P
	a) location of the adjustment means away from uninsulated hazardous live parts; or		P
	b) a guard to reduce the likelihood that the TOOL will contact with uninsulated live parts.		P

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Clause	Requirement + Test	Result - Remark	Verdict
	<i>Compliance is checked by inspection and, where necessary, by fault simulation.</i>		P
	Moving parts		N/A
	Moving parts that can cause injury to persons during service operations shall be located or protected so that unintentional contact with the moving parts is not likely.		N/A
	Capacitor banks		N/A
	Capacitor banks shall be fitted with a means of discharge for protection of SERVICE PERSONS. A warning label shall be added if discharge time exceeds 10 s, stating the time taken to reduce the hazard to a safe level (not greater than 5 min) (see 1.2.8.5/RD and 1.2.8.8/RD).		N/A
	Internal batteries		N/A
	Internal batteries shall be so arranged as to minimize risk of electric shock from inadvertent contact with terminals, and the interconnection method shall be such as to minimize risk of short-circuiting and electric shock during servicing and replacement. See 1.7.13.105.		N/A
	Compliance with 2.8.101.5 to 2.8.101.8 is checked by inspection.		N/A
	Electrical insulation		P
	Properties of insulating materials		P
	Additional subclauses:		P
	For PLUGGABLE EQUIPMENT TYPE B or permanently connected equipment intended to be installed by SERVICE PERSONNEL only, and marked as noted in 2.9.1.102, the humidity test in 2.9.2/RD is optional.		P
	Equipment on which the humidity test has not been performed shall be labelled "Intended for a CONTROLLED ENVIRONMENT". See 1.7.2.104 (q) for instruction manual requirements.		P
3	Wiring, connections, and supply		P
	<i>The provisions of 3/RD apply, together with the following:</i>		P
	General		P
3.1.4	Insulation of conductors		P
	<i>Additional subclause:</i>		P
	3.1.4.101 Insulated conductors of different circuits within a UPS, including wires in a terminal box or compartment, shall be either separated by barriers or segregated and shall be so separated or segregated from uninsulated live parts connected to different circuits.		P
3.2.5	Power supply cords		N/A
3.2.5.101	Cord-connected power supplies		N/A
	Power supplies intended to be cord-connected shall be provided with a suitable length of cord		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	having an additional conductor for grounding non-current-carrying conductive parts. The cord shall have an ampacity equal to or greater than the marked input in amperes, and shall be of the hard-usage type except:		
	a) as required by other clauses of this Standard;		N/A
	b) a cord for a table-mounted power supply having a mass of 5 kg or less may be Type SV (or equivalent) provided that the cord is not longer than 1 m.		N/A
	Disconnection from the mains supply		N/A
	Disconnect device – General		N/A
	A disconnect device shall be provided for the d.c. supply circuits and a.c. power output circuits of a permanently connected UPS and/or a remote supply/cabinet assembly. This disconnect device shall:		N/A
	a) open all ungrounded conductors;		N/A
	b) consist of either a manually operated switch or circuit breaker; and		N/A
	c) employ an operating handle that is either accessible from outside of the enclosure or located under a hinged cover not requiring a tool for opening.		N/A
	Constructions without additional disconnect means		N/A
	The following constructions may be provided without additional disconnect means:		N/A
	a) a UPS or battery supply having either a cord-and-plug or receptacle for connection to the output a.c. or d.c. circuit; and		N/A
	b) a permanently connected UPS or battery supply provided with an instruction manual indicating that the disconnection means is to be provided by others and the function of the disconnect device marked for its function. See Annex LLL .		N/A
	Disconnect device installed in a restricted access location		N/A
	A disconnect device installed in a restricted access location as defined in 1.2.7.3/RD is considered accessible and shall either		N/A
	a) be marked "This equipment is only to be installed either in a restricted area or an information technology equipment room"; or		N/A
	b) have the equivalent cautionary statement included in the installation manual.		N/A
	<i>Compliance is checked by inspection.</i>		N/A
	Interconnection of equipment		N/A
3.5.101	Telecommunication connectors		N/A
	Telecommunication-type connectors and terminals not intended for connection to the telecommunication network may be marked "Not for telecommunication (telephone) network" or an		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	equivalent wording, or with the symbol shown in Annex CCC. The marking shall be located in a readily visible location adjacent to the connector.		
4	Physical requirements		P
	<i>The provisions of 4/RD apply, together with the following:</i>		P
	Stability		P
	<i>Additional paragraph:</i>		P
	This test shall be applied to the unit with and without batteries.		P
4.3	Design and construction		P
4.3.8	Batteries		P
	Battery requirements		P
4.3.8.101.0	General		P
	Batteries shall comply with the following:		P
	a)UL 1973, the Standard for Batteries for Use in Stationary and Motive Auxiliary Power Applications; or		P
	b)UL 1989, the Standard for Standby Batteries, for lead acid batteries for UPS less than 70 kWhs.		P
	Location		P
	Vented batteries shall be contained in a separate enclosure or compartment from the power conversion portion of the UPS. This requirement does not apply to valve-regulated batteries and other sealed cell battery types. Vented batteries may be contained in the overall enclosure of the UPS, provided that a barrier, partition, or the equivalent is used to inhibit the passage of hydrogen gas from the battery area to the area of the enclosure containing the electronic components, switches, contactors, and controls used in the power conversion portion of the UPS.		P
	General requirements for installation		P
	Accessibility and maintainability		P
	Battery poles (terminals) and battery connectors shall be accessible so that their fixings (hardware) can be tightened with the correct TOOLS. Batteries with liquid electrolyte shall be so located that the battery cell (vent) caps are accessible for electrolyte tests and readjusting of electrolyte levels.		P
	<i>Compliance is checked by inspection and application of the TOOLS and measuring equipment supplied or recommended by the battery manufacturer.</i>		P
	Insulation		P
	Batteries in conductive casings shall be provided with adequate insulation between each other and from cabinets or compartments. Such insulation		P

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Clause	Requirement + Test	Result - Remark	Verdict
	shall meet the requirements of 2.9/RD.		
	<i>Compliance is checked by measurement and/or test.</i>		P
	Electrolyte spillage		P
	VENTED BATTERY trays and cabinets shall have an electrolyte-resistive coating.		P
	The ENCLOSURE or compartment housing a VENTED BATTERY shall be constructed so that spillage or leakage of the electrolyte from one battery will be contained within the ENCLOSURE and be prevented from		P
	a)reaching the outer surfaces of the UPS, where contact with the USER is possible;		P
	b)contaminating adjacent electrical components or materials; and		P
	c)bridging required electrical distances.		P
	<i>Compliance is checked by inspection.</i>		P
	Ventilation		P
	The ENCLOSURE or compartment housing a VENTED BATTERY shall be provided with proper ventilation so that any potentially explosive mixtures of hydrogen and oxygen are dispersed safely below non-hazardous levels.		P
	For battery compartments (separate or combined), the method for determining the necessary airflow to ensure adequate dissolution levels is given in Annex HHH.		P
	n combined apparatus of battery and electrical components, attention shall be given to prevent ignition of local concentrations of hydrogen and oxygen by adjacent operational arcing parts, such as contactors and switches close to battery vents/valves. This shall be achieved by the use of fully enclosed components, separation of battery compartments, or adequate ventilation, depending on the technical construction of the UPS and battery.		P
	For battery rooms, proper information on the required flow of air shall be provided in the installation instructions if the battery installation is supplied with the UPS.		P
	<i>Compliance is checked by inspection, calculation, and measurement.</i>		P
	Charging voltages		P
	Batteries shall be protected against excessive charging voltages under any normal or single fault condition.		P
	<i>Compliance is checked by the tests described in 5.3/RD and 5.3.102 of this Standard.</i>		P
	Thermal requirements		P
4.5.3	Temperature limits for materials		P
4.5.3.101A	Temperature limits		P

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Clause	Requirement + Test	Result - Remark	Verdict
	<i>Additional subclause:</i>		P
	4.5.3.101 For UPS with limited STORED ENERGY MODE duration and no provisions for additional batteries, the steady state temperature rise limits shown in Table 4.5.3.101A may be exceeded if the magnetic components meet the following criteria:		P
	a)the magnetic components operate during STORED ENERGY MODE only or operate during other modes but are likely to attain higher temperature rise during STORED ENERGY MODE operation, and duration of STORED ENERGY MODE operation does not exceed 60 min and the recharge time exceeds 8 times the STORED ENERGY MODE duration; and		P
	b)compliance is checked by conducting STORED ENERGY MODE operation immediately following the attainment of steady state temperatures during NORMAL LOAD operation. The maximum temperature rise of the windings measured immediately after expiration of stored energy duration shall not exceed the limits shown in Table 4.5.3.101B.		P
4.5.3.101B	Permitted temperature limits for magnetic windings at the end of stored energy mode operation		P
	Openings in enclosures		P
	Top and side openings		P
	Openings vertically above bare parts at hazardous voltages in the top of a FIRE ENCLOSURE or an electrical ENCLOSURE shall not exceed 5 mm in any dimension unless the construction prevents vertical access to such parts, for example, by means of a tray or similar restriction (see Figure 4B/RD). This requirement does not apply to equipment having openings in the top of an ENCLOSURE with a height exceeding 1.8 m. Requirements to prevent accessibility to hazardous parts apply.		P
5	Electrical requirements and simulated abnormal conditions		P
	<i>The provisions of 5/RD apply, together with the following:</i>		P
5.101	Short-circuit fault current withstand rating markings		P
	All UPS equipment may be marked with a short circuit withstand rating of 5 kA without any construction evaluation or testing. UPS equipment may be marked with a short circuit withstand rating greater than 5 kA up to 10 kA if it complies with the relevant provisions in Annex NNN. UPS equipment may be marked with a short circuit withstand rating greater than 10 kA up to 200 kA if it complies with the relevant provisions in Annex NNN.		P
	Electric strength		P
5.2.101	Electric strength test		P
	The terminals of across-the-line capacitors shall		P

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Clause	Requirement + Test	Result - Remark	Verdict
	withstand without breakdown 1000 volts plus twice rated a.c. voltage, or d.c. equivalent, applied for 1 min. Only capacitors not investigated to the standards listed in Annex III are subject to this test.		
	<i>An additional pulse current test is under consideration.</i>		P
	Abnormal operating and fault conditions		P
	Forced ventilation test		
	A UPS having forced ventilation shall be operated in all modes with the rotor of a blower motor or fan locked. For a UPS having more than one blower motor or fan, the test shall be conducted with the rotor of each blower motor or fan locked, either one at a time or simultaneously.		P
	A UPS having filters over ventilation openings shall be operated with the openings blocked to represent clogged filters. The test shall be conducted initially with the ventilation openings blocked approximately 50%, and then shall be repeated under fully blocked condition. A single blower or fan with a filter need not be tested under the fully blocked condition.		P
	The ventilating means for an ENCLOSURE or a compartment housing a battery shall comply with the requirements in HHH.2, HHH.4, and HHH.5 under blocked blower or fan and clogged filter conditions described in 5.3.101.1 and 5.3.101.2.		P
	Overcharge test		P
	A battery supply that is to be evaluated with the UPS shall be subjected to 7 h of overcharging, connected to a supply circuit adjusted to 106% of nominal voltage, using a fully charged battery. Any OPERATOR- adjustable controls associated with the charger or charging circuit shall be adjusted for the most severe charging rate.		P
	This requirement does not apply to a UPS provided with a regulating circuit preventing an increase in battery charging current and voltage when the a.c. input voltage is increased from rated value to 106% of rated value.		P
	This test can be waived for batteries that comply with the requirements for valve regulated and vented lead-acid and nickel-cadmium batteries in Annex H of UL 1973, the Standard for Batteries for Use in Stationary and Motive Auxiliary Power Applications.		P
	Output short-circuit test		P
	The direct current battery circuit and the alternating current output circuit of the UPS shall be shorted separately. For the alternating current output circuit, this test shall be conducted in all modes of operation that allow power to be delivered through the UPS. Shorting shall include from line to neutral (if provided) and from line to line.		P
	A battery supply is considered to comply with this		P

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Clause	Requirement + Test	Result - Remark	Verdict
	requirement if		
	a)An overcurrent-protection device is employed having a short-circuit interrupting rating not less than the maximum fault current available from the battery supply (this measurement shall be taken at the output of the device, i.e., output terminals or output cable ends, as applicable); and		P
	b)The maximum current from the battery supply during the STORED ENERGY MODE does not exceed the ampacity rating of the conductors connected to the batteries.		P
	With reference to the direct current battery circuit mentioned in 5.3.103.1, the test shall be conducted on		P
	a)the battery circuit of a UPS having an integral battery supply;		P
	b)the direct current output of a UPS intended to be used with a remote battery supply; and		P
	c)the direct current power circuit of a remote battery supply/cabinet assembly for use with a UPS and investigated under the requirements of this Standard.		P
	During the tests described in items a) and c), batteries intended to be used with the UPS shall be used and shall be fully charged. During the test described in item b), batteries shall not be connected. The tests mentioned in items b) and c) may be combined into one test with the batteries connected. See Figure 5.3.103.2.		P
5.3.103.2	Output short-circuit tests		P
	A unit having an integral battery supply in accordance with 5.3.103.2 (a)		P
	A unit intended to be used with a remote battery supply in accordance with 5.3.103.2 (b)		P
	A remote battery/supply cabinet in accordance with 5.3.103.2 (c)		P
	Abnormal test		P
	If an abnormal test is terminated by operation of the intended BRANCH CIRCUIT overcurrent protective device or if the a.c. input overcurrent protection is relied upon for protection of an a.c. output receptacle, the instruction manual for such a unit shall include the word "CAUTION" and appropriate safety wording.		P
	Output overload test		P
	This test is to be conducted after thermal stabilization is reached during the mode of operation, which allows the output power to be delivered through the power conversion portion of the UPS. While delivering maximum rated output power to an adjustable resistive load connected to the output ac circuit, the UPS is to be subjected to the overload test described in 5.3.105.2.		P
	The a.c. load is to be increased in increments of 10 % of the maximum output rating of the UPS and held for 1/2 h at each increment until:		P

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Clause	Requirement + Test	Result - Remark	Verdict
	a)Further change as a result of the test condition is not likely; or		P
	b)The UPS transfers to the BYPASS mode.		P
	Alternatively, thermal stabilization may be obtained with a load adjusted to result in maximum obtainable output power without causing operation of overcurrent protective devices, followed by increased incremental loading as described above.		P
	If the output power of a unit is not delivered through the power conversion portion of the unit during the normal mode of operation, the incremental loading is to be conducted in the reserve mode, following temperature stabilization in the NORMAL MODE.		P
6	Connection to telecommunication networks		N/A
	<i>The provisions of 6/RD apply.</i>		N/A
7	Connection to cable distribution systems		N/A
	<i>The provisions of 7/RD apply</i>		N/A
Annexes	All Annexes of the RD apply, together with the following:		N/A
Annex AAA (normative)	Additional component requirements		N/A
	The following component test requirements are in addition to those noted in 5.3/RD. These tests are applied one at a time with the equipment operating at rated input voltage, STORED ENERGY MODE, BYPASS mode, or other mode of operation as required by the specific test. Unless otherwise noted, the test criteria of 5.3/RD apply.		N/A
AAA.1	Capacitor surge current test		N/A
AAA.1.1	With the input a.c. power to the equipment disconnected, the battery supply fully charged, and the inverter storage capacitors discharged, the battery supply voltage shall be applied to the inverter input circuit by closing the battery supply disconnect switch intended to be used with the equipment. This test is not required for equipment:		N/A
	a)not having a manually operated battery supply disconnect switch; or		N/A
	b)having a battery supply disconnect switch that will not close onto the discharged capacitors without a.c. input power applied to the equipment.		N/A
AAA.1.2	A d.c. power source other than batteries may be used, provided that the source is capable of delivering an inrush current not less than that available from a fully charged battery supply.		N/A
AAA.2	Oil-filled capacitor fault test		N/A
AAA.2.1	In equipment having a bottom-ventilated ENCLOSURE containing oil-filled capacitors, the capacitors shall comply with the relevant		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	requirements of UL 810 and CSA C22.2 No. 190.		
AAA.2.2	This test is not required for equipment that complies with 4.7/RD		N/A
AAA.3	Electrolytic capacitor fault test		N/A
	For a UPS having d.c. electrolytic inverter storage capacitors, a fault in one of the capacitors in the inverter storage capacitor bank shall be simulated. This shall be accomplished by connecting the capacitor under test in reverse while the input supply to the equipment is not energized. The equipment shall then be energized. If the inverter is functional, then rated output load shall be applied.		N/A
AAA.4	Control devices		N/A
AAA.4.1	If the circuit controlled has a POWER FACTOR of less than 75%, the switch or other control device shall:		N/A
	a) have a horsepower rating based on the full-load ampere equivalent rating;		N/A
	b) have a current rating of not less than twice the rated input current; or		N/A
	c) withstand the overload and endurance tests specified in AAA.4.2 and AAA.4.3 .		N/A
AAA.1	Horsepower rating versus pilot duty rating		N/A
AAA.4.2	Overload (control devices)		N/A
AAA.4.2.1	A control device, except as permitted by AAA.4.2.6 , supplied as part of the equipment shall be capable of making and breaking for 50 cycles of operation at intervals of 10 s a current equal to 150% of the maximum load current at the actual POWER FACTOR involved.		N/A
AAA.4.2.2	The current-interrupting test of AAA.4.2.1 shall be made at the nominal supply voltage. The open-circuit voltage of the supply circuit shall be not less than 100%, or more than 150%, of the test voltage, except that a higher voltage may be employed if agreeable to those concerned. The current-carrying capacity of the supply circuit shall be such that the closed-circuit voltage with RATED CURRENT flowing is within $\pm 2.5\%$ of the test.		N/A
AAA.4.2.3	The ENCLOSURE of equipment designed for use on a system having one conductor grounded shall be connected during the test through a 3 A fuse to the grounded conductor of the circuit. If a power supply is designed for use on other types of systems, the frame shall be connected through such a fuse to the live pole least likely to arc to ground.		N/A
AAA.4.2.4	The test cycle shall be 1 s ON and 9 s OFF, if the nature of the switch permits the test to be so made.		N/A
AAA.4.2.5	There shall be no electrical or mechanical failure of the switch and no undue pitting, burning, or welding of the contacts, and there shall be no flashover to the ENCLOSURE.		N/A
AAA.4.2.6	As an alternative to the overload test specified in AAA.4.2.1, a switch or relay used to control an		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	internal load may be tested by making and breaking for 50 cycles of operation at the rate of 6 N/A cycles per minute the load that it is intended to control, with the equipment operation at 110% of the nominal supply.		
AAA.4.3	Endurance (control devices)		N/A
AAA.4.3.1	A control device supplied as part of the equipment shall be capable of making and breaking for 6000 cycles of operation at intervals of 10 s, a current equal to the maximum load current at the actual POWER FACTOR involved. There shall be no electrical or mechanical failure of the switch and no undue pitting, burning, or welding of the contacts.		N/A
AAA.4.3.2	The conditions for the endurance test shall be the same as the conditions for the overload test as indicated in AAA.4.2.2 to AAA.4.2.4.		N/A
AAA.5	Static transfer switches		N/A
AAA.5.1	General		N/A
AAA.5.1.1	Failure or malfunctioning of a solid state power-switching component shall result in a fail-safe condition (see AAA.5.3.2) under either of the following conditions:		N/A
	a) failure of a solid state power-switching component and another solid-state component such as in a control circuit; or		N/A
	b) failure of a solid state power-switching component and a mechanical component (e.g., relay, contactor, or the like) for static switches employing such a device in conjunction with a power-switching semiconductor. Failure analysis of the mechanical device shall not be required if the current through the contacts is the same or less than the RATED CURRENT of the device and the overload test evaluation is performed at six times RATED CURRENT. The control circuit that operates the mechanical device shall be subjected to a component failure analysis.		N/A
AAA.5.1.2	If a relay is used to transfer from the a.c. input line to the inverter, the overload current rating of the relay contacts shall be greater than the peak current that flows through the relay contacts during the output short-circuit test of 5.3.103 . The overload current rating shall be considered to be 150% of the rated load current for devices rated in A and 600% of the rated load current for devices rated in horsepower.		N/A
AAA.5.1.3	Overcurrent protection shall be provided for each static switch source in each ungrounded line. If the overcurrent devices are not integral with the equipment, they shall be specified on a label.		N/A
AAA.5.2	Isolating means		N/A
AAA.5.2.1	Mechanical isolation means shall be provided for servicing the power supply where failure of or leakage through a solid-state device could result in transfer of energy between two or more power		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	sources. The disconnecting means shall be provided as an integral part of the equipment and be connected into the circuit in such a way that when opened it will prevent transfer of energy between the different power sources. To facilitate servicing, barriers (or other suitable means) and/or marking shall be provided for protecting SERVICE PERSONNEL from live parts located ahead of isolating switches.		
AAA.5.2.2	Automatic mechanical isolating means shall be provided as an integral part of the equipment to prevent transfer of energy from a single-branch private supply system (such as a battery and inverter) to a branch of a multibranch system (such as a PRIMARY POWER system) upon loss or shutdown of the PRIMARY POWER system unless BACKFEED is not possible when evaluated in accordance with AAA.5.1.1 . The isolating means may be a relay, contactor, motorized circuit breaker, or other device with an air gap between contacts.		N/A
AAA.5.3	Tests		N/A
AAA.5.3.1	Static transfer switches shall be tested as part of the equipment to determine their acceptability for the application.		N/A
AAA.5.3.2	Compliance with the requirements of AAA.5.1.1 shall be verified by shorting one power switching component (e.g., SCR) and placing one fault on another component, such as in the control circuit. The fault placed on the component shall simulate the failure mode of that component (e.g., cathode to anode short on SCR, emitter to collector short or open on a transistor, or welding of a relay contact in the control circuit). This analysis shall be performed first with the PRIMARY POWER supply energized and repeated with the PRIMARY POWER supply not energized.		N/A
AAA.5.3.3	The electric strength test shall be performed on a static transfer switch with the solid-state devices, or assemblies, normally connected between the systems disconnected.		N/A
6	Transfer switches		N/A
AAA.6.1	N/A		N/A
AAA.6.2	A BYPASS switch or maintenance BYPASS used to connect the protected load directly to the BYPASS source shall comply with the requirements of AAA.6.3 .		
AAA.6.3	A BYPASS switch or maintenance BYPASS, when tested in conjunction with a UPS, shall comply with the load transfer test specified in AAA.7 and the applicable requirements of this Standard.		
AAA.7	Load transfer		
AAA.7.1	For an output transfer switch to be tested in accordance with AAA.6.3 , the BYPASS a.c. source shall be displaced from the a.c. output of the		N/A

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	equipment by:		
	a)120 electrical degrees for a 3-phase supply; or		N/A
	b)180 electrical degrees for a single-phase supply.		N/A
	The transfer switch shall be subjected to one operation of switching the load from the a.c. output of the equipment to a BYPASS a.c. source with the load adjusted to draw maximum rated a.c. power.		N/A
AAA.7.2	For equipment employing a BYPASS switch having a control to prevent switching between two a.c. sources out of synchronization, the test shall be conducted under the condition of a component failure if such a condition can result in an out-of-phase transfer between the two a.c. sources of supply. The fault placed on the component shall simulate the failure mode of that component (e.g., cathode to anode short on SCR, emitter to collector short or open on a transistor, or welding of a relay contact in the control circuit).		N/A
AAA.7.3	A solid-state BYPASS switch shall continue to operate normally after completion of the test described in AAA.7.1.		N/A
AAA.8	Bus bars		N/A
AAA.8.1	Each bus bar shall be plated at each joint with tin, silver, or nickel unless		N/A
	a)the joint is welded or brazed;		N/A
	b)the current at the joint is 600 A or less; or		N/A
	c)a suitable oxide-inhibiting compound is used over the joint surfaces.		N/A
	Other coatings may be used for aluminum bus bars if investigated for the application in accordance with the requirements for current-carrying parts described in AAA.8.13.		N/A
	AAA.8.2 The bending of a bus bar shall not result in visible cracks, but roughening or slight surface crazing shall be acceptable.		N/A
	AAA.8.3 Each riveted joint connection shall have a spring washer at one end and either a spring washer or a flat washer at the other end (see AAA.8.5 and AAA.8.6), unless, as an alternative, the connection is investigated in accordance with the applicable requirements of AAA.8.13.		N/A
	Washers may be omitted in a connection rated 225 A or less employing only copper bus bars.		N/A
	AAA.8.4 Each bolted joint connection shall employ a spring washer at one end (see AAA.8.6), unless the connection is investigated in accordance with the applicable requirements of AAA.8.13.		N/A
	A spring washer may be replaced with a split ring lock washer and flat washer if each bus in the joint is copper or if each aluminum bus in the joint has a tensile yield strength of at least 20 000 psi (138		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	MPa).		
	A flat washer, a split-ring lock washer, or a bolthead that complies with AAA.8.5 may be used in place of a spring washer if the joint does not include any aluminum or if aluminum bolts are used with aluminum bus bars.		N/A
	AAA.8.5 The flat washer referenced in AAA.8.3 and AAA.8.4 shall have a thickness of at least one-sixth that of the diameter of the rivet shank or bolt and shall have an outer diameter at least 150% of the rivet shank or bolt and not less than the outer diameter of any adjacent spring washer.		N/A
	AAA.8.6 The spring washer referenced in AAA.8.3 and AAA.8.4 shall be a dished washer of stainless or hardened and tempered steel, having an outer diameter not less than 150% of the bolt diameter, a thickness not less than one-eighth of the bolt diameter, and dished not less than 3-1/2% of the bolt diameter.		N/A
	AAA.8.7 Unless investigated for such use, a bolted connection between two bus bars or between a bus bar and another current-carrying part shall not depend on the dimensional integrity of a thermoplastic material.		N/A
	AAA.8.8 Insulation over bus bars, such as tape or tubing, shall not be provided over a bolted joint, so that tightening of the joint can be accomplished without removal of the insulation.		N/A
	AAA.8.9 The current density of a bus bar shall not be more than that indicated in Table AAA.8.1 or Table AAA.8.2 unless		N/A
	a)it has characteristics that will not result in maximum bus bar temperatures exceeding the values specified in Table AAA.8.3; or		N/A
	b)it is contained in a unit having forced air ventilation that will not result in maximum bus bar temperatures exceeding the values specified in Table AAA.8.3.		N/A
AAA.8.10	The cross-section of the bus bar referenced in Table AAA.8.1 or Table AAA.8.2 may be reduced by not more than 5% due to rounding, shaping, or dimensional tolerances.		N/A
AAA.8.11	Part of the bus bar material may be removed for slots or holes (whether used or not), provided that the remaining material at any cross-section along the length of the bus bar has at least 70% of the required ampacity in accordance with Table AAA.8.1 or Table AAA.8.2 and AAA.8.10, and provided that:		N/A
	a)the remaining metal in any 6 in (152 mm) length of bus is at least 93% of the metal of a bus having the required ampacity in accordance with Table AAA.8.1 or Table AAA.8.2 and AAA.8.10. For example, a 1 in (25.4 mm) wide bus could have 9/32 in (7.1 mm) holes on 1 in centres, or a 4 in		N/A

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	(102 mm) wide bus could have 13/32 in (10.3 mm) wide slots 3.2 in (81.3 mm) long every 6 in (152 mm); or		
	b)the bus bar has characteristics that will not result in temperatures exceeding the values specified in Table AAA.8.3.		N/A
AAA.8.12	The limitations of current density mentioned in Table AAA.8.1 and Table AAA.8.2 do not apply to		N/A
	a)a connecting strap, bus, or the like comprising a part of a circuit breaker, switch, or fuseholder employed in the unit; and		N/A
	b)a portion of a strap, bus, jumper, or the like adjacent and connected to a terminal of a switch, circuit breaker, fuse, or fuseholder, but not more than 1 in (25.4 mm) from the terminal, if a reduced cross-section in that portion is necessary because of the recessing of the terminal or because of barriers adjacent to it.		N/A
AAA.8.13	Bus bar tests		N/A
AAA.8.13.1	An aluminum bus bar employing a coating in accordance with AAA.8.1 or a bus bar that has the alternative clamped joint construction permitted by AAA.8.3 and AAA.8.4 shall be subjected to the tests described in AAA.8.13.2 to AAA.8.13.4.		N/A
AAA.8.13.2	The temperature of the bus bar joint shall be measured during the temperature test described in 4.5/RD and shall comply with the maximum temperature specified in Table AAA.8.3.		N/A
AAA.8.13.3	The temperature rise at the joint during the five hundredth cycle shall not be more than 15 °C (27°F) higher than the temperature rise at the end of the 25th cycle.		N/A
AAA.8.13.4	The test sample shall consist of an assembly of bus bars connected together to form a series circuit. The bus bars shall be clamped together with the joint construction used in actual production. The number and size of the bus bar shall represent the maximum ampere rating and the maximum current density in which the joint construction is employed. This may necessitate more than one test. The length of each bus bar shall be 2 ft (609 mm). The bus bar shall be connected to a power supply by any convenient means that will not affect the joint temperature. The power supply shall be adjusted to deliver a value of current that will result in a temperature of 75 °C (135 °F) above room temperature at the joint. The assembly shall then be subjected to a 500 cycle test. At the end of the 24th cycle, the current shall be readjusted to bring the temperature of the joint to 75 °C (135 °F) above room temperature, and this current value shall be maintained for the remainder of the cycling test. At the end of the 25th and 500th cycle, the		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	temperatures shall be recorded. The temperatures shall be measured on both sides of the joint as close as possible to the bolt or rivet. The cycling rate shall be 3 h ON and 1 h OFF. The ON period during which temperatures are recorded may be extended to more than 3 h if necessary for the joint to attain thermal equilibrium.		
Annex BBB (informative)	Examples of NORMAL LOAD conditions		N/A
BBB.1	General		N/A
BBB.1.1	The UPS is loaded according to the manufacturer's specifications given in the instruction manual.		N/A
	If the specifications are missing, the following NORMAL LOAD conditions can be used.		N/A
	A UPS can be loaded with different linear and NON-LINEAR LOADS.		N/A
	A LINEAR LOAD is defined by the fact that with a sinusoidal voltage supplied to such a load, the current will be sinusoidal also.		N/A
	A NON-LINEAR load with sinusoidal voltage has non-sinusoidal current.		N/A
BBB.1.2	The most common types of LINEAR LOADS are:		N/A
	a)resistive;		N/A
	b)inductive-resistive; and		N/A
	c)capacitive-resistive.		N/A
BBB.1.3	A NON-LINEAR LOAD can be:		N/A
	a)rectified capacitive load; or		N/A
	b)thyristor or transducer controlled load (phase control)		N/A
BBB.1.4	In the low power range below 3 kVA, the rectifier in bridge connection with capacitive load is the most common.		N/A
BBB.2	For resistive loads, the UPS is loaded with a resistor (RL) up to nominal power.		N/A
BBB.3	For inductive-resistive loads, an inductance is connected in series or in parallel with a resistor.		N/A
	The resistor (R) and inductance (L) are given by the following formulas:		N/A
	a)series connection		N/A
	b)parallel connection		N/A
BBB.4	For capacitive-resistive loads, a capacitance and a resistor are connected either in series or in parallel.		N/A
	The resistor (R) and capacitance (C) are given by the following formulas:		N/A
	a)series connection		N/A
	b)parallel connection		N/A
BBB.5	To simulate a rectified capacitive load, the UPS is loaded with a diode rectifier bridge, which has a		N/A

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	capacitor and a resistor in parallel on its output.		
	A ripple voltage, 5% peak to peak of the capacitor voltage U_c , corresponds to a time constant of $R_1 \times C = 0.15 \text{ s}$.		N/A
	Observing peak voltage, distortion of line voltage, voltage drop in line cables, and ripple voltage of rectified voltage, the average of the rectified voltage U_c will be as follows: $U_c = 2 \times 0.92 \times 0.96 \times 0.975 \times U = 1.22 \times U$ and the values of resistors R_s , R_1 , and capacitor C will be as follows: $R_s = 0.04 \times U^2 / S$ $R_1 = U^2 / (0.66 \times S)$ $C = 0.15 \text{ s} / R_1$ The voltage drop in the diode bridge is neglected.		N/A
Annex CCC (informative)	Safety symbols and their meanings		N/A
Annex DDD (normative)	Harmonic distortion test		N/A
DDD.1	When tested as described in DDD.3 , the total rms of the harmonic voltages, excluding the fundamental, delivered by a UPS shall not exceed 30% of the fundamental rms output voltage, and the rms voltage of any single harmonic shall not exceed 15% of the fundamental rms output voltage. The specified output voltage distortion levels may be exceeded if the UPS complies with DDD.4.		N/A
DDD.2	If the inverter portion of the UPS does not operate to supply output during the normal mode of operation, the UPS shall be operating in STORED ENERGY MODE during this test.		N/A
DDD.3	With reference to the requirements in DDD.1 , the UPS shall be connected to an adjustable resistive load at rated power. The measurements shall be made at open circuit and with the UPS delivering 25, 50, 75, and 100% of rated power.		N/A
DDD.4.1	The tests described in DDD.4.2 and DDD.4.3 shall be conducted to investigate the acceptability of output voltage harmonic distortion exceeding the levels specified in DDD.3 .		N/A
DDD.4.2	The load shall be energized from the following sources, one at a time:		N/A
	a) the a.c. output of the UPS while the output is supplied by the inverter portion of the UPS; and		N/A
	b) a PRIMARY POWER source of sinusoidal voltage adjusted to the same rms value as obtained from the output of the UPS inverter. While energized from each source, temperatures of components that may be affected by harmonic voltages shall be measured. Such components include, but are not limited to, 1) linear transformer winding; 2) d.c. filter capacitor in a switch-mode power		N/A

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	supply; 3)across-the-line capacitor in an electromagnetic interference (EMI) filter; 4)continuous duty capacitor start motor winding; and continuous duty shaded pole motor winding.		
	The maximum temperatures obtained while the load is supplied from the UPS shall not exceed by more than 5 °C (9 °F) the corresponding maximum temperatures obtained when the load is supplied from the PRIMARY POWER source.		N/A
DDD.4.3	If the output voltage distortion of the unit varies with the amount of load connected to the UPS, the condition of loading resulting in the highest level of distortion shall be used for this test. To accomplish this, it may be necessary to connect other loads in parallel with the load under test.		
Annex EEE (normative)	Earthing and bonding		N/A
EEE.1	General requirements		N/A
EEE.1.1	An output a.c. power circuit shall be grounded if the circuit:		N/A
	a)has no electrical connection to supply conductors originating in another wiring system;		N/A
	b)a PRIMARY POWER source of sinusoidal voltage adjusted to the same rms value as obtained from the output of the UPS inverter. While energized from each source, temperatures of components that may be affected by harmonic voltages shall be measured. Such components include, but are not limited to,		N/A
	1)linear transformer winding;		N/A
	2)d.c. filter capacitor in a switch-mode power supply		N/A
	3)across-the-line capacitor in an electromagnetic interference (EMI) filter;		N/A
	4)ontinuous duty capacitor start motor winding; and		N/A
	5)continuous duty shaded pole motor winding.		N/A
	The maximum temperatures obtained while the load is supplied from the UPS shall not exceed by more than 5 °C (9 °F) the corresponding maximum temperatures obtained when the load is supplied from the PRIMARY POWER source.		N/A
DDD.4.3	If the output voltage distortion of the unit varies with the amount of load connected to the UPS, the condition of loading resulting in the highest level of distortion shall be used for this test. To accomplish this, it may be necessary to connect other loads in parallel with the load under test.		N/A
Annex EEE (normative)	Earthing and bonding		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
e)			
EEE.1	General requirements		N/A
EEE.1.1	An output a.c. power circuit shall be grounded if the circuit:		N/A
	a)has no electrical connection to supply conductors originating in another wiring system;		N/A
	b)is rated 50 to 600 V; and		N/A
	c)is as follows (other configurations may require grounding):		N/A
	1)a circuit that is grounded so that the maximum voltage to ground on the ungrounded conductors does not exceed 150 V. This requires that one conductor of each of the following circuits does not exceed 150 V and is grounded:		N/A
	i)120 volts, 2-wire;		N/A
	ii)240/120 volts, single-phase, 3-wire;		N/A
	iii)208/120 volts, two-phase, 3-wire; and		N/A
	iv)208/120 volts, three-phase, 4-wire.		N/A
	2)a circuit nominally rated 480 wye/277 V, 3-phase, 4-wire, in which the neutral is used as a circuit conductor; and		N/A
	3)a circuit nominally rated 240/120 V, 3-phase, 4-wire, in which the midpoint of one phase is used as a circuit conductor.		N/A
	Reference should be made to local electrical codes for other circuits.		N/A
EEE.1.2	The conductor to be grounded shall be as follows:		N/A
	a)single-phase a.c. system, 2-wire: one conductor;		N/A
	b)single-phase a.c. system, 3-wire: the neutral conductor;		N/A
	c)multiphase a.c. system having one wire common to all phases: the common conductor; and		N/A
	d)multiphase a.c. system where one phase is used as a single-phase, 3-wire, a.c. system: the neutral conductor.		N/A
EEE.1.3	Grounding of the circuits shall be made by a bonding jumper connected between the grounded conductor and to:		N/A
	a)the ENCLOSURE of a metal-enclosed unit; or		N/A
	b)the metal chassis that is bonded to the equipment's protective earthing means of a nonmetallic enclosed unit.		N/A
	The size of the bonding jumper shall be as specified in column 4 of Table EEE.1.		N/A
EEE.1.4	The circuit may be grounded in the field if the following provisions are implemented:		N/A
	a)A field-wiring terminal is connected to the circuit by a bonding jumper. The size of the terminal and jumper shall not be less than that specified in column 4 of Table EEE.1.		N/A
	b)The field-wiring terminal is identified as the neutral by a metallic-plated coating substantially		N/A

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	white in colour that is readily distinguishable from the other terminals, or proper identification of the terminal, such as a marking on the unit, or an indication on a wiring diagram attached to the unit, or information provided in the instruction manual; and		
	c)the following marking is provided identifying the circuit as a separately derived source: "The output a.c. circuit is considered as a separately derived source. If local codes require grounding of this circuit, use terminal '_____' for bonding this circuit to the ENCLOSURE. Ground the ENCLOSURE to a suitable grounding electrode in accordance with local code requirements. Refer to the instruction manual".		N/A
EEE.1.5	A fixed unit shall be provided with a terminal that complies with EEE.2 for connection of the grounding electrode conductor to the metal ENCLOSURE or equipment grounding conductor. The terminal shall be:		N/A
	a)capable of securing a conductor of a size as specified in column 3 of Table EEE.1; and		N/A
	b)marked with the following words: "Grounding Electrode Terminal".		N/A
	For an a.c. output circuit of a unit having a polarized receptacle, lead, or terminal identified as a grounded circuit that is not grounded at the unit itself because of an electrical connection to supply conductors originating in another wiring system, a potential involving risk of electric shock shall not exist between ground and the grounded circuit contact, terminal, or lead.		N/A
EEE.1.6	Compliance with the requirements of EEE.1 shall be determined by neutral-to-ground potential measurement. The potential shall not exceed 30 V rms (42.4 V peak) at no-load and full-load conditions.		N/A
EEE.2	Wiring terminals		N/A
EEE.2.1	Equipment field-wiring terminals shall be marked as follows:		N/A
	a)"Use Copper Conductors Only" if the terminal is acceptable only for connections to copper wire;		N/A
	b)"Use Aluminum Conductors Only" or "Use Aluminum or Copper-Clad Aluminum Conductors Only" if the terminal is acceptable only for connection to aluminum wire; and		N/A
	c)"Use Copper or Aluminum Conductors" or "Use Copper, Copper-Clad Aluminum, or Aluminum Conductors" if the terminal is acceptable for connection to either copper or aluminum wire.		N/A
EEE.2.2	Except as provided in EEE.2.3, a wiring terminal shall be provided with a pressure terminal connector that is securely fastened in place; for example, firmly bolted or held by a screw (not a crimping type).		N/A
EEE.2.3	A wiring terminal may be provided with a pressure terminal connector that is not securely fastened in		N/A

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	place if the construction is one of the following:		
	a)A wire-binding screw may be employed at a wiring terminal intended for connection of a 10 AWG (5.3 mm ²) or smaller conductor if upturned lugs, a cupped washer, or the equivalent is provided to hold the wire in position.		N/A
	b)A pressure terminal connector, including a crimping type, may be field-installed if all of the following conditions are met:		N/A
	1)The connector assemblies are available from the unit manufacturer and are specified in the instruction manual.		N/A
	2)The fastening hardware required for an effective installation is either provided with the unit or specified in the instruction manual.		N/A
	3)The installation of the connector shall not involve the loosening or disassembly of parts other than a cover. The means for securing the connector shall be readily accessible for tightening before and after the installation of conductors.		N/A
	4)Instructions are provided identifying the TOOL and procedure for installing the connector if other than an ordinary TOOL is required.		N/A
	5)Installation of the connector in its intended manner will not result in a risk of hazard as defined by this Standard.		N/A
EEE.2.4	A wiring terminal shall be prevented from turning or shifting in position by a means other than friction between surfaces. This may be accomplished by two screws or rivets; square shoulders or mortises; a dowel pin, lug, or offset; a connecting strap or clip fitted into an adjacent part; or an equivalent method.		N/A
EEE.2.5	A wiring terminal that secures the wire by crimping need not be prevented from turning if the required creepage and CLEARANCE distances are maintained with the terminal oriented in the position resulting in the least spacing between adjacent terminals or between the terminal and conductive parts.		N/A
EEE.2.6	A wire-binding screw at a field-wiring terminal shall not be smaller than No. 10 (4.8 mm diameter head) unless		N/A
	a)a No. 8 (4.2 mm diameter head) screw is used at a terminal intended only for the connection of a 14 AWG (2.1 mm ²) conductor or a 16 or 18 AWG (1.3 or 0.82 mm ²) SELV CIRCUIT conductor; or		N/A
	b)a No. 6 (3.5 mm ² diameter head) screw is used for the connection of a 16 or 18 AWG (1.3 or 0.82 mm ²) SELV CIRCUIT conductor.		N/A
	A wire-binding screw shall thread into metal.		N/A
	A terminal plate tapped for a wire-binding screw shall be of metal with a minimum thickness of 0.050 in (1.27 mm).		N/A
	A terminal plate tapped for a wire-binding screw may be of metal with a minimum thickness less than 0.050 in (1.27 mm) if the terminal plate is used		N/A

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	for a SELV or limited energy circuit and the tapped threads have adequate mechanical strength to withstand the tightening torque test specified in Table EEE.3.		
	There shall be two or more full threads in the metal of a terminal plate. The metal may be extruded at the tapped hole to provide at least two full threads.		N/A
	Two or more full threads are not required in the metal of a terminal plate if the terminal plate is used for a SELV or limited energy circuit and the tapped threads have adequate mechanical strength to withstand the tightening torque test specified in Table EEE.3.		N/A
	EEE.2.7 A terminal for connection of a grounded conductor of an a.c. power circuit shall be identified as the neutral by one of the following methods:		N/A
	a)a metallic-plated coating substantially white in colour that is readily distinguishable from the other terminals;		N/A
	b)proper identification of the terminal, such as a marking on the unit, or an indication on a wiring diagram attached to the unit, or information provided in the instruction manual; or		N/A
	c)if field wiring leads are provided, the lead intended to be grounded shall have a white or gray colour and shall be readily distinguishable from other leads.		N/A
Annex FFF (normative)	BACKFEED PROTECTION test		N/A
FFF.1	General		N/A
	A UPS shall not allow excessive touch currents to be available between any pairs of input terminals of the UPS during its STORED ENERGY MODE of operation. Where the measured open circuit voltage does not exceed 30 V rms (42.4 V peak, 60 V d.c.), the touch current measurement need not be taken.		N/A
	The BACKFEED PROTECTION for a stationary UPS, with an output that is not a separately derived system, shall open or de-energize all supply conductors.		N/A
	Compliance is checked by the tests in described in FFF.2 and FFF.3.		N/A
FFF.2	Pluggable type A UPS		N/A
	While the UPS is in its STORED ENERGY MODE of operation, and with the input terminals or plugs disconnected, the following conditions shall apply for both no-load and full-load conditions:		N/A
	a)Under normal and any single fault conditions, the current shall not exceed 3.5 mA when measured between any two USER-accessible input terminals, using the circuit shown in Annex D/RD.		N/A
	b)If such protection is provided by an internal system, such a system shall operate within 1 s of		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	the disconnection of the input terminals.		
FFF.3	Permanently connected UPS		N/A
	The test and conditions shall be as for FFF.2, except the protective conductor shall not be disconnected during the test, the current shall not exceed 3.5 mA for both no-fault and single-fault conditions, and the protection shall operate within 15 s.		N/A
	When BACKFEED PROTECTION is provided by others and requires a remote shunt-trip or other means of mechanical disconnection, adequate interconnect information shall be included in the installation instructions.		N/A
FFF.4	Single fault conditions		N/A
	For FFF.2 and FFF.3, single-fault conditions shall be determined by circuit inspection and/or investigation, but shall also include potential load faults such as phase-to-earth isolation failures.		N/A
	For single-phase output:		N/A
	For three-phase output:		N/A
FFF.5	Air gap requirements for mechanical disconnect		N/A
	An air gap is only required when the BACKFEED PROTECTION is mechanical in nature. The air gap is defined as the CLEARANCE distance in accordance with the RD. There are several elements to consider when determining the CLEARANCE requirement:		N/A
	a) Under normal operation, the space between poles of phases shall meet the requirements for BASIC INSULATION. Reference Tables 2M/RD and 2N/RD.		N/A
	b) If the unit is operating on inverter, the source is considered to be a secondary supply, which is transient free. Reference Table 2M/RD (last column). For example, circuits less than 150 V rms require 0.7 mm for BASIC INSULATION, and circuits greater than 150 V rms but less than 300 V rms require 1.4 mm for BASIC INSULATION. For a UPS with floating outputs, opening all phases and the neutral using the required CLEARANCE for BASIC INSULATION is considered acceptable. If the output is grounded to the chassis, REINFORCED INSULATION or equivalent is required.		N/A
	c) CLEARANCE of components meeting an applicable component standard may be further reduced, provided that they meet both the manufacturing quality control program that has at least the same level of assurance as the example given in Annex R.2/RD and the withstand voltage of Table G.2/RD. For example, 0.4 mm is an acceptable air gap for voltages not exceeding 300 V rms.		N/A
FFF.6	ELECTRONIC BACKFEED PROTECTION		N/A

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	If the mechanism for BACKFEED PROTECTION relies on electronic controls only, the components of the mechanism shall be able to withstand the effects of transient overvoltages, voltage variations, electromagnetic susceptibility, electrostatic discharge, shipping, storage, thermal cycling, and humidity. See also Annex LLL.		N/A
Annex GGG (informative)	Principles of BACKFEED PROTECTION		N/A
GGG.1	General		N/A
GGG.1.1	UNINTERRUPTIBLE POWER SYSTEMS store and generate hazardous energy. These energies may be present at the input terminals of the UPS.		N/A
	BACKFEED PROTECTION is intended to prevent USERS, SERVICE PERSONNEL, or electricians from unforeseeable or unnecessary exposure to such hazards. The control scheme is called BACKFEED PROTECTION. BACKFEED PROTECTION functions when a.c. input power is removed and hazardous energies could otherwise be present at the input terminals of the UPS.		N/A
	Two levels of protection are assumed:		N/A
	a)Level 1: OPERATOR ACCESS AREA – BACKFEED PROTECTION is required where USER exposure is likely. An example of this would be any PLUGGABLE EQUIPMENT TYPE A UPS on which the USER could unplug the unit and touch the pins of the line cord plug or appliance coupler of the UPS.		N/A
	b)Level 2: RESTRICTED ACCESS LOCATION – BACKFEED PROTECTION is required where SERVICE PERSONNEL or electrician exposure is likely, but USER exposure is not likely. An example of this would be a PLUGGABLE EQUIPMENT TYPE B or permanently connected UPS on which the only access to BACKFEED would be internal to the unit or in the service panel supplying power to the UPS.		N/A
	The USER is protected within one second from disconnection for PLUGGABLE EQUIPMENT TYPE A devices, and SERVICE PERSONNEL and electricians must be protected within 15 s for PLUGGABLE EQUIPMENT TYPE B or permanently connected devices. This time includes any time necessary to discharge filter capacitors.		N/A
	BACKFEED PROTECTION can be mechanical or electronic.		N/A
	Mechanical BACKFEED PROTECTION should meet a minimum air gap requirement. If not, the mechanical device (contacts) may be forced closed, and this will not be counted as a fault.		N/A
	ELECTRONIC BACKFEED PROTECTION includes a mechanism to determine that it is functioning properly. This mechanism is intended to		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	be single-fault-tolerant. The BACKFEED PROTECTION operates with any and all semiconductor devices in any single phase of the PRIMARY POWER path failed.		
	Where the UPS is supplied with more than one a.c. source, such as in a BYPASS condition, the BACKFEED PROTECTION will disconnect all primary source connection points from the internal hazardous energy of the UPS.		N/A
	BACKFEED PROTECTION works under any normal operating condition. This should include any output load or input source condition deemed normal by the manufacturer; however, it is common practice to only test at full- and no-load conditions, unless analysis of the circuitry proves other conditions would be less favourable. The circuitry that controls the BACKFEED PROTECTION is intended to be single-fault-tolerant.		N/A
	BACKFEED PROTECTION can accomplish this by disconnecting the AC MAINS SUPPLY wiring from the internal energy source, by disabling the inverter and removing the hazardous source(s) of energy, reducing the source to a safe level, or by placing a suitable mechanical barrier between the USER, SERVICE PERSONNEL, or electrician and the hazardous energy.		N/A
GGG.1.2	Hazardous energy is defined in the body of this Standard. The method of measurement is as follows:		N/A
	a)For pluggable equipment, it is determined by opening all phases, neutral and ground.		N/A
	b)For permanently connected devices, the neutral and ground are not removed during the BACKFEED tests.		N/A
GGG.1.3	Measurements are taken at the UPS input connections across the phases, from phase to neutral and phase and neutral to ground, using the body impedance model as the measurement device.		N/A
GGG.2	Air gap requirements for mechanical disconnect		N/A
	An air gap is only required when the BACKFEED PROTECTION is mechanical in nature. The air gap is defined as the CLEARANCE distance in accordance with the RD. There are several elements to consider when determining the CLEARANCE requirement:		N/A
	a)Under normal operation, the space between poles of phases must meet the requirements for BASIC INSULATION. Reference Tables 2M/RD and 2N/RD.		N/A
	b)If the unit is operating on inverter, the source is considered to be a secondary supply, which is transient free. Reference Table 2M/RD (last column). For example, circuits less than 150 V rms require 0.7 mm for BASIC INSULATION, and circuits greater than 150 V rms but less than 300 V rms require 1.4 mm for BASIC INSULATION. For a		N/A

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	UPS with floating outputs, opening all phases and the neutral using the required CLEARANCE for BASIC INSULATION is considered acceptable. If the output is grounded to the chassis, REINFORCED INSULATION or equivalent is required.		
	CLEARANCE of components meeting an applicable component standard may be further reduced, provided that they meet both the manufacturing quality control program that has at least the same level of assurance as the example given in Annex R.2/RD and the withstand voltage of Table G.2/RD. For example, 0.4 mm is an acceptable air gap for voltages not exceeding 300 V rms.		N/A
GGG.3	Fault testing		N/A
	All BACKFEED PROTECTION control circuits are subject to failure analysis and testing.		N/A
GGG.4	Relays		N/A
	Relays in the power path that are required to open for BACKFEED PROTECTION should be normally open when not energized.		N/A
	If the relay does not meet the required CLEARANCES, the shorting of either pole/contact may be considered as a single fault to simulate the welding of the contacts. The failure of a single relay contact may be sensed and the inverter disabled to prevent feedback.		N/A
	The relay must be horsepower-rated or pass a 50 cycle endurance test at 600% of the normal switching current.		N/A
GGG.5	Electronic protection		N/A
	Electronic protection for BACKFEED PROTECTION is acceptable if the operation of the electronic protection device is sensed and the inverter is disabled if a fault is found. This is the same requirement as for a relay having less than the required air gap or CLEARANCE.		N/A
GGG.6	Mechanical protection		N/A
	Mechanical protection for BACKFEED PROTECTION is acceptable if it prevents the USER from accessing HAZARDOUS VOLTAGES and cannot be readily defeated without the use of a TOOL. The voltage rating of the mechanical protection should be no less than the maximum out-of-phase voltage.		N/A
GGG.7	Control circuitry		N/A
	The failure, open- or short-circuit, of any component of the BACKFEED PROTECTION circuitry may be analyzed to evaluate the effects on the proper operation of the BACKFEED PROTECTION. Testing may be done on all components where analysis of the results is arguable.		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	Components such as resistors and inductors are considered to fail open-circuit only. Capacitors may fail open or shorted. Solid-state devices typically fail short and then open.		N/A
	Microprocessor controls are considered to be acceptable if the circuit operates safely with any single control line open or shorted to control logic ground. Failure of the microprocessor can also be simulated by opening the Vcc pin or shorting the Vcc pin to ground.		N/A
	If the control circuitry is fully redundant, for example, N + 1, failure analysis of individual components is not required if the failure of one circuit results in a fail-safe mode of operation.		N/A
Annex HHH (normative)	Ventilation of battery compartments		N/A
HHH.1	The requirements of HHH.2 to HHH.5 apply to units having VENTED BATTERIES.		N/A
HHH.2	The ENCLOSURE or compartment housing a battery shall be vented where gassing is possible during heavy discharge, overcharging, or similar type of usage. The means of venting shall provide airflow throughout the compartment in order to reduce the risk of buildup of pressure or accumulation of a gas mixture, such as hydrogen-air, that involves a risk of injury to persons.		N/A
HHH.3	Arcing parts, such as the contacts of switches, circuit breakers, and relays, shall not be located in the ENCLOSURE or compartment housing a VENTED BATTERY described in HHH.2, nor shall the ENCLOSURE or compartment vent into closed spaces where such parts are located. For purposes of this requirement, fuses and connectors are not considered to contain arcing parts. Battery or compartment monitoring sensors (such as temperature sensors and the like) may be located in the ENCLOSURE or compartment.		N/A
HHH.4	If the gas mixture is lighter than air (such as hydrogen-air), the requirement of HHH.2 may necessitate locating additional ventilation openings in the uppermost portions of the battery ENCLOSURE or compartment where such a gas mixture may accumulate.		N/A
HHH.5	With reference to HHH.4, the venting means shall prevent hydrogen concentrations in excess of 1% by volume. If the adequacy of the ventilation required in HHH.2 is not obvious, a determination shall be made by measurement of gas concentration in accordance with the battery compartment ventilation test in HHH.7. A battery at full charge, when most of the charging energy goes into gas, will generate approximately 1 ft ³ (28.3 L) of hydrogen gas per cell for each 63 ampere-hours of input. See HHH.6.		N/A

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HHH.6	The ventilating means for an ENCLOSURE or a compartment housing a battery shall comply with the requirements of HHH.2 to HHH.5 under blocked fan and blocked filter conditions described in 5.3.101.		N/A
HHH.7	if a measurement is needed to determine if a battery compartment complies with HHH.5, the battery supply shall be subjected to the overcharge test in 4.3.8.101.2.5. During, and at the conclusion of the test, the maximum hydrogen gas concentration shall not be more than 2% by volume. Measurements shall be made by sampling the atmosphere inside the battery compartment at the periods of 2, 4, 6, and 7 h during the test. Samples of the atmosphere within the battery compartment shall be taken at the location where the greatest concentration of hydrogen gas is likely, using an aspirator bulb provided with the concentration measurement equipment, or other equivalent means.		N/A
HHH.7A	The gas generation determination conducted as part of the Overcharge Thermal Runaway Test of UL 1973, the Standard for Batteries for Use in Stationary and Motive Auxiliary Power Application, can be utilized to determine suitable ventilation for vented batteries instead of running the test of HHH.7.		N/A
HHH.8	In the United States the following applies:		N/A
	If the battery is located in a HAZARDOUS VOLTAGE, PRIMARY, or ELV CIRCUIT, the battery shall conform to the Standard for Batteries for Use in Stationary and Motive Auxiliary Power Application, UL 1973.		N/A
Annex III (normative)	Standards for components		N/A
Annex JJJ (normative)	French translations and markings		N/A
	This annex shall be used in conjunction with Annex NAA/RD and identifies examples of additional markings required for Canada and the United States.		N/A
	Excluding the words "WARNING" and "CAUTION", wording equivalent to that provided in this annex may be used.		N/A
	French translations of required markings are considered informative. It is the responsibility of the manufacturer to provide bilingual markings, where applicable, in accordance with local jurisdictional requirements.		N/A
Annex LLL (informative)	Examples of US and Canadian regulatory requirements		N/A

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ve)			
	This annex is used in conjunction with Annex NAE/RD and provides additional examples of and references for regulatory requirements that may apply to equipment. Applicability of these requirements is dependent on the construction of the equipment and its intended installation and use.		N/A
	This annex is not intended to provide a complete list of all of the applicable requirements, only to serve as a reference for requirements that most commonly apply to this type of equipment.		N/A
	For complete requirements, the <i>National Electrical Code</i> , ANSI/NFPA 70, the <i>Canadian Electrical Code, Part I</i> , CSA C22.1, or other referenced document must be consulted.		N/A
	Any undated reference to a code or standard appearing in the requirements of this Standard is to be interpreted as referring to the latest edition of that code or standard. N/A		N/A
Annex MMM (normative)	Test and construction requirements for UPS with fault current ratings higher than those specified in Table NNN.5		N/A
MMM.1	Short-circuit current		N/A
	The UPS shall have one or more short-circuit rating values as specified in Table MMM.1, but the value shall not be less than the values based on an equipment rating as specified in Table MMM.2.		N/A
MMM.2	Maximum 100 000 ampere short-circuit current rating without short-circuit test		N/A
MMM.2.1	Scope		N/A
	These requirements cover UPS switchboard sections and UPS sections that do not contain semiconductor switching components, and that have an rms symmetrical short-circuit current rating for which short-circuit tests may be waived. These requirements do not otherwise amend the requirements in this Standard.		N/A
MMM.2.2	Construction		N/A
MMM.2.2.1	General		N/A
MMM.2.2.1.1	A maximum rms symmetrical short-circuit current rating as shown in Table MMM.3 may be assigned to a through bus (which includes a splice bus), a tap, and a section bus, of a UPS rated as specified in Table MMM.3 without conducting short-circuit tests if all of the following conditions are met:		N/A
	a)The construction is in accordance with MMM.2.2.1.2 – MMM.2.2.1.8		N/A
	b)The performance is in accordance with MMM.2.3.1– MMM.2.3.3.		N/A
	c)The UPS is marked in accordance with MMM.2.4.		N/A
MMM.2.2.	For a maximum assigned rms symmetrical short-		N/A

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1.2	circuit current rating of 100 000 A, the UPS (consisting of one or more sections) shall contain a single integral main molded-case circuit breaker, low voltage a.c. power circuit breaker, or fused switch having a short-circuit rating not less than that of the UPS, or shall be marked for use with a remote device having a short-circuit current rating not less than that of the UPS. Supply bus bars ahead of the main overcurrent protective device shall be located in the same section or an adjacent section.		
MMM.2.2.1.3	Copper or aluminum bus bars shall be nominally 1/4 in (6.4 mm) thick and have a width as described in Table MMM.3; holes in bus bars shall not be larger than 0.438 × 0.813 in (11.1 × 20.7 mm) with the larger dimension limited to use along the axis of the bus bar.		N/A
	Larger holes may be provided as specified in MMM.2.2.1.4 for provision of current transformers.		N/A
MMM.2.2.1.4	With regard to Figure MMM.2, copper or aluminum bus bars shall be minimum 1/4 in (6.4 mm) thick and 2 to 7 in (50.8 to 178 mm) wide; holes in bus bars at SUPPORTS shall not be larger than 0.406 × 0.750 in (10.3 × 19.1 mm) and holes for bus bar type current transformers shall not be larger than 9/16 × 1 13/32 in (14.3 × 35.7 mm). A bus bar shall be prevented from rotating by means other than the mounting bolt specified in MMM.2.3.3.		N/A
MMM.2.2.1.5	Bus bar configurations shall be flat and rectangular.		N/A
	An L-shaped bus bar may be used if		N/A
	a) it is considered to be a face-to-face configuration; or		N/A
	b) the unsupported leg does not extend more than 2 in (50.8 mm) from the bus SUPPORT.		N/A
MMM.2.2.1.6	The number of bus bars shall not exceed four for each phase.		N/A
MMM.2.2.1.7	Phase bus bars, including neutral bus bars, shall be arranged edge-to-edge or face-to-face as described in Table MMM.3 and as shown in Figure MMM.1– Figure MMM.9.		N/A
	A neutral bus bar need not comply with this requirement if the spacing between the adjacent surface of the neutral bus bar and the nearest phase bus bar is		N/A
	a) at least 5-1/4 in (133 mm) or		N/A
	b) 5 in (127 mm) if the short-circuit current rating of the UPS does not exceed 50 000 A.		N/A
MMM.2.2.1.8	Stacked bus bars shall not exceed a height of 1-3/4 in (44.5 mm) measured from the top of the standoff insulator or nonmetallic channel base to the top of the highest bus bar.		N/A
	At a splice bus or a joint, the height of stacked bus bars shall not exceed 2 in (50.8 mm)		N/A
	The height may be greater than 1-3/4 in for		N/A

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	constructions using L-shaped bus bars as specified in MMM.2.2.1.5.		
MMM.2.2.2	Spacings		N/A
MMM.2.2.2.1	The spacing between adjacent edges of bus bars of opposite polarity, arranged or mounted edge-to-edge as shown in Figure MMM.8, shall not be less than as described in Table MMM.3. The center-to-center distance shall also be not less than as shown in Table MMM.3. The distance between bolts to standoff insulators or nonmetallic channels along the bus bar shall comply with the dimensions specified in Table MMM.3. Bus bars to which terminal connectors are secured shall extend not more than 4 in (102 mm) from the center of the first bolt to a standoff insulator on a nonmetallic channel.		N/A
MMM.2.2.2.2	The center-to-center spacing between adjacent groups of bus bars of opposite polarity, arranged or mounted face-to-face as shown in Figure MMM.9, shall not be less than as described in Table MMM.3. The distance between closest points of opposite polarity shall also be not less than as shown in Table MMM.3.		N/A
MMM.2.2.2.3	The spacing between bus bars crossing at right angles as shown in Figure MMM.2, Figure MMM.3, Figure MMM.5, Figure MMM.6, and Figure MMM.7 shall not be less than 1 in (25.4 mm).		N/A
MMM.2.2.2.4	The spacing between live parts of bus bars mounted or arranged edge-to-edge and grounded, non-current carrying metal parts shall not be less than 1-3/8 in (34.9 mm) in any direction. A barrier shall be disregarded when measuring this spacing.		N/A
	The spacing between the flat face of an insulated neutral bus bar and grounded, non-current carrying metal parts shall not be less than 1/2 in (12.7 mm) for a short-circuit current rating of 42 000 A or less.		N/A
MMM.2.2.2.5	The spacing between live parts of bus bars mounted or arranged face-to-face and grounded, non-current carrying metal parts, measured perpendicular to the face of the bus bar, shall not be less than 6 in (152 mm); and, measured perpendicular to the edge of the bus bar, shall not be less than 1-3/8 in (34.9 mm). A barrier shall be disregarded when measuring this spacing.		N/A
	The spacing between the flat face of a neutral bus bar measured perpendicular to the face of the bus bar and grounded, non-current carrying metal parts shall not be less than 2 in (50.8 mm).		N/A
	The spacing-to-grounded metal may be 1-3/8 in minimum within 3 in (76.2 mm) of the bus bar SUPPORT.		N/A
MMM.2.2.3	SUPPORTS		N/A
MMM.2.2.3.1	A standoff insulator SUPPORT used for direct support of live parts shall be secured by a bolt of		N/A

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	the type described in MMM.2.3.3 to a steel channel having the dimensions shown in Figure MMM.10. The bolt shall pass through a clearance hole in the steel channel and shall not thread into it. Typical constructions are shown in Figure MMM.11 and Figure MMM.12.		
	A non-metallic channel that is investigated in accordance with MMM.2.3.1 – MMM.2.3.3 may be used in place of the standoff insulator and steel channel.		N/A
MMM.2.2.3.2	A bus bar shall be secured to a standoff insulator, a nonmetallic channel, or a splice bus bar by a bolt of the type described in MMM.2.3.3.		N/A
MMM.2.2.3.3	A bolt used to secure a bus bar or standoff insulator shall be 3/8-16 steel or larger and shall be torqued to 20 lbf (27.1 N·m). A steel washer with an outside diameter of 1 in (25.4 mm) or larger and a maximum inside diameter of 9/16 in (14.3 mm) shall be located between the bolt head and the bus bar and also between the bolt head or nut and a steel or nonmetallic channel. The washer shall be minimum 0.078 in (2.0 mm) thick. A bolt or threaded rod shall be SAE grade 5 minimum.		N/A
MMM.2.2.3.4	The spacing and number of bus SUPPORTS shall be in accordance with Figure MMM.1 – Figure MMM.7 as applicable.		N/A
MMM.2.2.3.5	The line terminal BRACE shown in Figure MMM.1 shall be secured to the line terminal bus bars by bolts of the type described in MMM.2.3.3.		N/A
	The line terminal BRACE is not required in a construction having bussed line connections.		N/A
MMM.2.3	Performance		N/A
MMM.2.3.1	General		N/A
	A standoff insulator or nonmetallic channel shall be subjected to the tests described in MMM.2.3.2 and MMM.2.3.3. Three samples shall be subjected to each test. The results shall be acceptable if the standoff insulator or nonmetallic channel does not crack.		N/A
MMM.2.3.2	Tensile		N/A
	The bus bar SUPPORT or representative sample shall be subjected to a minimum tensile force of 1850 lb (8229 N) applied between the simulated enclosure supporting means and bus bar SUPPORT along the centerline of the mounting bolts. See Figure MMM.13 for typical constructions. The force shall be applied by machine parts moving apart at a rate of 10 in (254 mm) per minute.		N/A
MMM.2.3.3	Cantilever		N/A
	The standoff insulator or nonmetallic channel or representative sample shall be mounted to a		N/A

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	simulated enclosure steel plate and subjected to a minimum force of 675 lb (3003 N), applied as specified and as shown in Figure MMM.14. A 4-in (102 mm) length of 2 × 1/4-in (50.8 × 6.4 mm) copper bus bar shall be mounted to the SUPPORT at a distance of 1-1/2 in (38.1 mm) from the standoff insulator or nonmetallic channel. Single or multiple spacers not greater than 1 × 1 in (25.4 × 25.4 mm) are to be used. The force shall be applied perpendicular to the 2 by 1/4 in edge of the bus bar. The force shall be applied by parts of a machine moving apart at a rate of 10 in (254 mm) per minute.		
MMM.2.4	Marking		N/A
	The UPS shall be marked with the following or the equivalent: "Wrap line cables together and, if provided, tap cables together with nominal 3/8-in (9.5 mm) nylon rope or rope having a minimum tensile strength of 2000 lb (8896 N) at 6 in (152 mm) and 12 in (305 mm) from the line terminals with five wraps and, every additional 6 in with five wraps or every 1 in (25.4 mm) with one wrap." It is also recommended that a drawing as shown in Figure MMM.15, or the equivalent, be provided with the UPS.		N/A
Annex NNN (normative)	Short-circuit withstand rating		N/A
NNN.1	Rating verification and marking		N/A
	If a UPS is marked with a SHORT-CIRCUIT WITHSTAND RATING higher than 5 kA, then the manufacturer shall verify its rating through short-circuit tests in Annex NNN or construction methods specified in Annex MMM. A marked SHORT-CIRCUIT WITHSTAND RATING shall be in accordance with Table NNN.1 or Table NNN.2.		N/A
NNN.2	Short-time withstand current		N/A
NNN.2.1	General		N/A
	The short-time withstand current test shall be performed as a type test to verify the safety of the UPS when a short circuit is applied across the output terminals under prescribed modes of operation, except where exempted. Refer to Figure NNN.1 for a typical circuit for implementation of the test and to NNN.3.1.1 for exemptions.		N/A
NNN.2.2	Modes of operation		N/A
	Testing is performed only in modes of operation wherein the output power is delivered by the a.c. input through a low impedance path.		N/A
NNN.3	Test procedure		N/A
NNN.3.1	General application		N/A
	The UPS a.c. input shall be connected to a supply		N/A

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	capable of delivering the prospective test current in accordance with Table NNN.1. The UPS shall be in the appropriate mode of operation (see NNN.2.2) and otherwise operating without load and at rated input voltage and frequency. A short-circuit shall then be applied across the output terminals of the UPS. A UPS rated for multiple input and output voltages may be tested at any of its rated input voltages provided that applicable interrupting components have been certified or tested to interrupt the prospective test current at the highest rated input voltage.		
	The test shall be repeated by placing a short circuit across the phase closest to the neutral terminal when the latter is provided. However, the phase-to-neutral test is not required when the neutral construction is at least as robust as that of the phase conductors in terms of cross-sectional area, mechanical support and clearance.		N/A
	If the manufacturer declares a short-time withstand current higher than shown in Table NNN.1, the declared value applies for test purposes.		N/A
	The test is considered complete when the prospective test current has been made available for the minimum duration listed in Table NNN.1.		N/A
NNN.3.1.1	Exemption from testing		N/A
	Exemption from short-time withstand current testing applies to:		N/A
	a)UPS marked with a short-circuit withstand rating of 5 kA or less;		N/A
	b)UPS marked with a short-time withstand rating equal to or less than the lowest rated current-limiting protective device in the UPS that does not exceed 10 kA;		N/A
	c)UPS intended to be supplied from transformers whose rated power does not exceed 10 kVA per phase for a rated secondary voltage of not less than 110 V, or 1,6 kVA per phase for a rated secondary voltage less than 110 V, and whose short-circuit impedance is not less than 4 %; or		N/A
	d)UPS models with nameplate ratings of equal or lower capacity and identical construction to a tested model; Note: The exempt models must be marked with a short-time withstand rating equal to or less than the rating of the tested model.		N/A
	Compliance is verified by satisfying at least one of the exemption conditions.		N/A
NNN.3.1.2	Schematic of typical test setup		N/A
	Figure NNN.1 shows a typical schematic of the short-circuit test set up. The schematic within the dotted block represents a typical UPS construction.		N/A
NNN.3.1.3	Testing port requirements		N/A
	A UPS rated for multiple input and output voltages shall be tested at the highest rated input voltages provided that applicable interrupting components have been evaluated to interrupt the prospective		N/A

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	test current at the highest rated input voltage. For a UPS with multiple rated withstand markings, the port indicated in Figure NNN.2 shall be tested. The following two tests are required at each port being marked on the UPS with a withstand rating:		
	a)Short-circuit making capacity "Closing test"; and		N/A
	b)Short-circuit capacity "Withstand test".		N/A
NNN.3.2	Sample selection		N/A
NNN.3.2.1	Test provisions		N/A
	Representative samples shall be tested to determine the performance of each principal bus configuration or cable arrangement, or both. If provision is made for line and load cables, the sample selection and test program shall include cables of the minimum cross-sectional area that would be used for field connection for each size of wire connector used. The combination of these factors can result in more than one test being conducted.		N/A
NNN.3.2.2	Choosing representative samples		N/A
	In choosing representative samples, the following factors shall be considered:		N/A
	a)Length of current-carrying members or cables;		N/A
	b)Material and cross-sectional configuration of each bus bar structure;		N/A
	c)Weakest bus bar structure that could result in bus bar distortion;		N/A
	d)Stiffest bus bar structure most likely to transmit the maximum forces to the bus SUPPORT;		N/A
	e)SUPPORTs or bracing structures that represent the weakest mechanical supports;		N/A
	f)Components and similar features;		N/A
	g)Venting of a circuit breaker toward or near a live part or a grounded metal part (shortest uninsulated electrical spacing);		N/A
	h)Blocking or partial blocking of a vent in a circuit breaker;		N/A
	i)Mounting of a circuit breaker where its vents are directly opposite the vents in another circuit breaker; and		N/A
	j)Location that will cause a maximum let through current in the circuit breaker under test.		N/A
	Upon short-circuit testing a sample with a particular bus bar or bus bar SUPPORT configuration, testing of an alternate configuration shall not be necessary as long as the alternate has the same physical shape and a material with the same or greater mechanical strength.		N/A
NNN.3.2.3	Relative strength of parts		N/A
	When determining the relative strength of hinges, latches, screws, and enclosures, the following characteristics shall be compared:		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	a)material and thickness of parts;		N/A
	b)type and size of fasteners;		N/A
	c)diameter of hinge pins and construction of hinge joints;		N/A
	d)number of hinges, latches, and screws;		N/A
	e)distance between hinges, latches, and screws; and		N/A
	f)size of door or cover.		N/A
NNN.3.3	Sample preparation		
NNN.3.3.1	UPS installation		N/A
	The sample shall be mounted and supported as in a normal installation. All unused openings other than ventilation openings, such as holes for conduit, shall be closed.		N/A
	The doors of the UPS, if provided, shall be closed and latched in the intended manner during the test. A door, cover, or filler plate shall be installed over any unused branch circuit device space. All covers shall be in place during the test.		N/A
	A cover may be omitted if the cover does not give access to a circuit breaker included in the test circuit and if the test does not result in movement of live parts toward the location of the covers that would result in a reduction of spacings.		N/A
NNN.3.3.2	Enclosure fuse		N/A
	if the UPS intended to be used on input sources has one conductor grounded, it shall be tested with the enclosure(s) connected to the grounded conductor through a 30 A, non-delay-type cartridge fuse.		N/A
	If the UPS is intended for use on other types of input circuits, the enclosure shall be connected through the fuse to the live pole least likely to strike to the normally grounded parts.		N/A
	The fuse shall have a voltage rating not less than the test voltage. The connection shall be made to the load side of the limiting impedance, as shown in Figure NNN.1, by a No. 10 AWG (5.3 mm ²) copper wire having a total circuit length not exceeding 6 ft (1.83 m).		N/A
	The enclosure shall be insulated from the supporting floor.		N/A
	The optional switch (15A in Figure NNN.1) can be used to isolate UPS cabinets from the source so that the operator can turn the UPS and its controls ON prior to conducting the short-circuit test. If this switch is used, it shall be closed prior to performing short-circuit tests.		
NNN.3.3.3	Circuit breakers		N/A
	A circuit breaker having adjustable trip features shall have all adjustments set at maximum current and time settings.		N/A

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NNN.3.4	Line (input) connections		N/A
NNN.3.4.1	Connection of UPS inputs		N/A
	If the UPS is provided with separate inputs for rectifier and alternate (BYPASS) source, both inputs may be connected to single short-circuit current test source.		N/A
NNN.3.4.2	Input conductor size		N/A
	The line terminals of the UPS (8 in Figure NNN.1) shall be connected to the source terminals (4 in Figure NNN.1) by a short cable run (5 in Figure NNN.1), having ampacity based on 75°C insulation, nearest to but not less than the input current rating of the UPS. The cables shall enter the UPS at the designated entry area at a point that provides maximum length of unsupported cables within the UPS enclosure. The cable terminals shall be wired and tightened to the torque as marked on the UPS. There shall be no bracing of the cables inside the enclosure unless information is included with the equipment. The cables may be braced as they leave the enclosure to the source side terminals (4 in Figure NNN.1).		N/A
NNN.3.4.3	Input conductor type		N/A
	The sample shall be tested with a copper conductor if the UPS is restricted for use with a copper conductor as marked on the equipment.		N/A
	The sample may be tested with an aluminum or copper conductor if the short-circuit current rating divided by the number of cables per phase results in a current of 50 000 A per cable, or less.		N/A
	A UPS that does not have a provision for wire connection shall be connected with bus bars of the same ampacity as the supply bus.		N/A
	If the short-circuit current rating is greater than 50 000 A per conductor, the test sample may be tested with either a compact aluminum or copper conductor if the type of conductor used for the short-circuit test has a lower pull-out force than the conductor intended for field installation.		N/A
NNN.3.4.4	Length of supply conductors		N/A
	The length of the supply conductors shall not exceed 8 ft (2.44 m) per terminal unless the excess length is included in the test circuit calibration as covered in NNN.3.7.		N/A
NNN.3.4.5	Integral fuseholders or fusible switches		N/A
	In a UPS provided with an integral main fusible switch, the supply conductor shall be connected to the terminals of the UPS. The test fuse mentioned in NNN.3.3.2 shall be installed in the main fusible switch. If the size of the test fuse is such that it cannot fit in the fuseholder, an external fuseholder shall be used. The external fuseholder shall be inserted either:		N/A
	a) between the load side of the fusible switch and the		N/A

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	main bus bar; or		
	b) the line side of the fusible switch.		N/A
NNN.3.4.6	Length of supply cable and leads		N/A
	The combined length of the supply cable and of all other leads shall not exceed 8 ft (2.44 m) per terminal unless the excess length is included in the test circuit calibration as covered in NNN.3.7.		N/A
NNN.3.4.7	Copper bus bar or tube		N/A
	If external fuses are used, a copper bus bar or tube (dummy fuse) shall be installed in each fuseholder of the main fusible switch. The copper bus bar, tube, or dummy fuse shall have a cross section not less than that of the blade or ferrule of the fuse that the fuseholder is intended to accommodate. Each bus bar or tube may be individually reinforced to enable it to withstand the short-circuit forces. A bus bar, tube, or dummy fuse shall be secured in place in the same manner as a fuse in intended service.		N/A
NNN.3.5	Load (output) connections		N/A
NNN.3.5.1	Output conductors		N/A
	The load terminals of the UPS (10 in Figure NNN.1) shall be connected to the short-circuiting switch (11 in Figure NNN.1) by conductors, having ampacity based on 75°C insulation, nearest to but not less than the current rating of the UPS output protective device internal or external to the UPS. The cables shall enter the UPS at the designated entry area at a point that will provide maximum length of unsupported cables within the UPS enclosure. The cable terminals shall be wired and tightened to the torque as marked on the UPS. There shall be no bracing of the cables inside the enclosure unless information is included with the equipment. The cables may be lashed outside the UPS to prevent whipping during the test.		
NNN.3.5.2	Length of output conductors		N/A
	The length of the load conductors shall not exceed 4 ft (1.22 m) per terminal outside of the enclosure of the UPS.		N/A
NNN.3.5.3	UPS with output circuit breaker		N/A
	If the UPS output circuit protective device is an internal or external circuit breaker having an adjustable trip feature, it shall have all adjustments set at maximum current and time settings		N/A
NNN.3.5.4	Output conductors from circuit breaker		N/A
	Cables having an ampacity based on 75°C insulation, nearest to but not less than the rating of the circuit breaker, shall be run from each circuit breaker terminal to the short circuiting switch located outside the UPS.		N/A
NNN.3.5.5	UPS with output fusible switch		N/A
	If the UPS output circuit protective device is a		N/A

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	fusible switch, the maximum ampere rated fuse that can be inserted shall be installed in the fuseholders, unless marked otherwise on the equipment. The test fuse mentioned in NNN.3.3.2 shall be installed in the fusible switch.		
	If the size of the test fuse is such that it cannot fit in the fuseholder, an external fuseholder shall be used and a copper bus bar or tube (dummy fuse) shall be installed in the fusible switch. The dummy fuse shall have a cross-section no less than that of the blade or ferrule of the fuse that the fuseholder is intended to accommodate. The dummy fuse may be reinforced to enable it to withstand the short-circuit forces. The dummy fuse shall be secured in place in the same manner as are the fuses in normal use.		N/A
NNN.3.5.6	Output conductors from fusible switch		N/A
	Cables having an ampacity based on 75°C insulation, nearest to but not less than the rating of the fusible switch, shall be run from each fusible switch terminal to the short circuiting switch located outside the UPS. If the test fuses are located outside the equipment, conductors having a length of no more than 4 ft (1.22 m) per terminal shall be used from each switch terminal to the test fuses.		N/A
NNN.3.6	Short-circuit test procedure		N/A
NNN.3.6.1	General requirements		N/A
	The following requirements apply for UPS equipment that is required to be short-circuit tested:		N/A
	a)equipment intended for use on an alternating-current system shall be tested with alternating-current at rated frequency as indicated in Figure NNN.1;		N/A
	b)three-phase equipment shall be tested on a 3-phase circuit;		N/A
	c)single-phase equipment, including a construction using adjacent poles of a 3-phase construction, shall be tested on a single-phase circuit;		N/A
	d)the open circuit voltage at the supply connection (4) shall be 100 to 105% of the rated maximum voltage of the equipment;		N/A
	e)the test supply frequency shall be within ± 2 Hz of the rated frequency;		N/A
	f)neutral short-circuit tests shall be made at the marked line-to-neutral voltage and short-circuit current rating of the equipment, using the neutral and nearest phase conductor;		N/A
	g)the available short-circuit rms symmetrical current in amperes, as determined in accordance with NNN.3.8, shall not be less than the rated and marked value and not less than the value specified in Table NNN.1;		N/A
	h)for equipment incorporating circuit breakers or fuses, the available short-circuit test current shall not be greater than the marked interruption current rating of the breakers or fuses;		N/A

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	i)the test source circuit shall include the necessary measuring equipment and the fuse-mounting means as required;		N/A
	j)the POWER FACTOR of the circuit shall be no higher than that specified in Table NNN.3; and		N/A
	k)controlled closing shall be used for all tests as specified in Table NNN.4.		N/A
NNN.3.6.2	Air core reactor shunt resistance		N/A
	The reactive components of the impedance in the line shown in Figure NNN.1 may be paralleled if of the air-core type. Reactance shall not be connected in parallel with resistance, except that an air-core reactor in any phase may be shunted by resistance (R) as calculated by the formula: $R = 167 \times (E / I)$ in which: E is the voltage across the air-core reactor with current I flowing as determined by oscillographic measurement during the short-circuit calibration or, by proportion, from meter measurements at some lower current.		N/A
NNN.3.6.3	Short-circuit test duration		N/A
	The duration of a short-circuit test shall be:		N/A
	a)the duration necessary to operate an integral or separate overcurrent device;		N/A
	b)1-1/2 cycles for equipment that has a short-circuit rating of 10 000 A, is marked in accordance with NNN.5.2, and is protected by a molded case circuit breaker rated 400 A or less from a specified or unspecified manufacturer;		N/A
	c)a duration equal to or greater than the maximum opening time of the specified overcurrent protective devices at the level of current involved during the short circuit current if the short circuit is performed without overcurrent protective devices as marked per NNN.5.2; or		N/A
	d)not less than three cycles if neither (a) nor (b) nor (c) applies.		N/A
NNN.3.7	Test circuit calibration		N/A
NNN.3.7.1	Measurement of test circuit characteristics		N/A
	An oscillograph may be used to measure the short-circuit test circuit characteristics in order to determine that when the system is short-circuited, the available current and circuit parameters (currents, voltages, POWER FACTOR, closing angle, and time duration) shall be in accordance with Table NNN.1, Table NNN.2, Table NNN.3, and Table NNN.4 as applicable. Calibration tests may be conducted at lower-than-rated current and voltage.		N/A
NNN.3.7.2	Calibration at UPS input terminals		N/A
	To determine the short-circuit test circuit for the available rms symmetrical current and POWER FACTOR, short circuit the line terminals of the		N/A

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	separate main device (6 in Figure NNN.1) or, if no separate main device is used, the line terminals of the UPS (8 in Figure NNN.1).		
NNN.3.7.3	Calibration at test station terminals		N/A
NNN.3.7.3.1	Determination of available current		N/A
	If the length of the cable run from test station terminals (4 in Figure NNN.1) and UPS input terminals (8 in Figure NNN.1) does not exceed 8 ft (2.44 m), then the available current shall be:		N/A
	a) the current at the test station terminals (4 in Figure MMM.1) without adjustment for circuits rated 25 000 A or less;		N/A
	b) 5% or greater than the required test current for circuits rated between 25 001 and 50 000 A; or		N/A
	c) 10% or greater than the required test current for circuits rated between 50 001 and 200 000 A.		N/A
NNN.3.7.3.2	Determination of test current and POWER FACTOR		N/A
	The magnitude of the test current and the POWER FACTOR of the circuit shall be determined by the applicable requirements of UL 489 (molded case circuit breakers, molded case switches, and circuit breaker enclosures) and CSA C22.2 No. 5 (molded case circuit breakers, molded case switches and enclosures for circuit breakers).		N/A
NNN.3.8	Short-circuit test		N/A
NNN.3.8.1	Recording test parameters		N/A
	The current waveforms, voltage waveforms, POWER FACTOR, closing angle, and time duration shall be recorded during the short-circuit current withstand test in NNN.3.8.2 and NNN.3.8.3.		N/A
NNN.3.8.2	Short-circuit test with rectifier circuit and inverter circuit		N/A
	The following steps shall be performed sequentially on the UPS equipment if the rectifier and inverter are not excluded from the short-circuit test (see NNN.2.2) :		N/A
	a) Open short-circuit current source switch (1) and remove all shorting connections used for calibration tests.		N/A
	b) Connect UPS terminals to source terminals and open switches (15A and 11).		N/A
	c) Close short-circuit source switch (1) and close the main protective device (7);		N/A
	d) Close UPS input circuit breakers (CB1, if provided, and CB2) and adjust the short-circuit test source voltage and frequency in accordance with NNN.3.6.1 (d).		N/A
	e) Turn ON the UPS and the static bypass switch, and operate at no-load.		N/A
	f) If the battery section is not excluded from the test, close the battery (CB) or disconnect.		N/A
	g) If output and isolation circuit breakers (CB3 and		N/A

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	CB5) are provided, close these devices.		
	h)Close isolation switch (15A).		N/A
	i)Close the short-circuiting switch (11) at the closing angle in accordance with NNN.3.6.1 (k).		N/A
	If the UPS does not transfer to the BYPASS during the test in step (i) or the bypass circuit does not carry short-circuit current, then repeat the short-circuit test in BYPASS mode in accordance with NNN.3.8.4.		N/A
NNN.3.8.3	Short-circuit test without rectifier circuit and inverter circuit		N/A
	The following procedure steps shall apply sequentially to the UPS if the rectifier and inverter are excluded from the short-circuit test. The UPS rectifier and inverter need not be energized or operating during this test:		N/A
	a)Open short-circuit current source switch (1), and remove all shorting connections used for calibration tests.		N/A
	b)Connect UPS alternate (BYPASS) input terminals to source terminals, and open switches (15A and 11).		N/A
	c)Close the short-circuit source switch (1) and the main protective device (7).		N/A
	d)Close UPS input circuit breakers (CB1, if provided, and CB2) and adjust the short-circuit test source voltage and frequency in accordance with NNN.3.6.1(d).		N/A
	e)Turn ON the UPS BYPASS static switch and operate at no-load.		N/A
	f)Close isolation circuit breaker (CB5), if provided.		N/A
	g)Close isolation switch (15A).		N/A
	h)Close the short-circuiting switch (11) at the closing angle in accordance with NNN.3.6.1(k).		N/A
NNN.3.8.4	Maintenance BYPASS section		N/A
NNN.3.8.4.1	Required tests		N/A
	A UPS provided with internal maintenance BYPASS breaker or switch shall be subjected to a closing test (see NNN.3.8.4.2), an opening test (see NNN.3.8.4.3), and a dielectric test (see NNN.3.8.4.4). The UPS rectifier, inverter, and BYPASS static switch may be de-energized or inoperative during these tests.		N/A
NNN.3.8.4.2	Closing test		N/A
	The following steps shall apply for the closing test:		N/A
	a)Close the maintenance BYPASS breaker or switch (CB4 in Figure NNN.1).		N/A
	b)Close the maintenance BYPASS breaker or switch (CB4 in Figure NNN.1).		N/A
	Close the short-circuiting switch (11) at the closing angle in accordance with NNN.3.6.1 (k).		N/A

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NNN.3.8.4.3	Opening test		N/A
	The following steps shall apply for the opening test:		N/A
	a)Open the maintenance BYPASS breaker or switch (CB4).		N/A
	b)Close the short-circuiting switch (11).		N/A
	c)Adjust the short-circuit test source voltage and frequency in accordance with NNN.3.6.1 (d).		N/A
	d)Close the maintenance BYPASS breaker or switch (CB4) at random.		N/A
NNN.3.8.4.4	Dielectric test		N/A
	Following the short-circuit current withstand test, the UPS equipment shall be subjected to the dielectric voltage-withstand (electric strength) test described in 5.2.		N/A
NNN.4	Compliance criteria		N/A
	After short-circuit testing, the UPS may be inoperative; however, the UPS shall meet the following conditions:		N/A
	a)no flames, burning particles, or debris emitted during or after the conclusion of the short-circuit test;		N/A
	b)no opening of the 30 A enclosure fuse (NNN.3.3.2);		N/A
	c)no evidence of arcing between live parts of opposite polarity;		N/A
	d)enclosure doors or covers remain operational;		N/A
	e)supplemental doors or covers over circuit breaker handles not opened more than 60° from the closed position;		N/A
	f)no live parts accessible when tested in accordance with 2.1.1.1/RD;		N/A
	g)no creepage and clearance distances reduced below required minimum values in the RD;		N/A
	h)no conductor pulled out of a terminal connector;		N/A
	i)no damage to the conductor or to the insulation of the conductor;		N/A
	j)no damage to component devices, such as circuit breakers or fuseholders, beyond that permitted by the applicable component standards;		N/A
	k)no damage to the plug-in or draw-out assemblies;		N/A
	l)no separation of a bus bar insulator, insulating SUPPORT, or cable restraint into two or more pieces;		N/A
	m)no cracks on opposite sides of a base and no cracks, including surface cracks, running the full length or width of the insulating SUPPORT; and		N/A
	n)the equipment, with all switching devices closed, withstands the electric strength test (see 5.2), which may be repeated with a semiconductor switching device removed if a breakdown occurs between that device and ground on the previous		N/A

UL 1778			
Clause	Requirement + Test	Result - Remark	Verdict
	test.		
NNN.5	Markings		N/A
NNN.5.1	UPS not requiring protection by a specific overcurrent device		N/A
	A UPS that does not require protection by a specific overcurrent device shall be marked as follows:		N/A
	"This UPS is rated for use on a circuit capable of delivering no more than [short-circuit current] rms symmetrical amperes, [voltage] volts maximum."		N/A
	The marked SHORT-CIRCUIT WITHSTAND RATING in the first blank line ("short-circuit current") shall be in accordance with the symmetrical values given in Table NNN.1 and Table NNN.2.		N/A
NNN.5.2	UPS dependent on use of a specific overcurrent device		N/A
	If the short-circuit current rating of the UPS is dependent upon the use of a specific overcurrent protective device ahead of the UPS, the UPS shall be marked as follows if a fuse is used for the protective device:		N/A
	"When protected by a [fuse manufacturer's name] [fuse type] type [fuse rating] ampere maximum class [fuse designation] fuse, this UPS is rated for use on a circuit capable of delivering no more than [short-circuit current] rms symmetrical amperes, [voltage] volts maximum."		N/A
	The fourth blank line ("fuse designation") shall be filled in with one of these designations: CC, G, J, L, RK- 1, RK-5, or T, unless the fuse being used is a semiconductor-type fuse complying with CSA C22.2 No. 248.13 or UL 248-13 and restricted to a specific part number of a specific manufacturer.		N/A
	The marked SHORT-CIRCUIT WITHSTAND RATING in the fifth blank line ("short-circuit current") shall be in accordance with the symmetrical values given in Table NNN.1 or Table NNN.2.		N/A
	Correspondingly, the UPS shall be marked as follows if a circuit breaker is used for the protective device:		N/A
	"When protected by a [circuit breaker manufacturer's name] [breaker type] type circuit breaker rated no more than [breaker rating] amperes, this UPS is rated for use on a circuit capable of delivering no more than [short-circuit current] rms symmetrical amperes, [voltage] volts maximum."		N/A
	The marked SHORT-CIRCUIT WITHSTAND RATING in the fourth blank line ("short-circuit current") shall be in accordance with the symmetrical values given in Table NNN.1 or Table NNN.2.		N/A
NNN.5.3	UPS dependent on use of a current-limiting device		N/A

UL 1778			
Clause	Requirement + Test	Result - Remark	Verdict
	If the short-circuit current rating of the UPS is dependent upon the use of a current-limiting device upstream of the UPS, the UPS shall be marked as follows:		N/A
	"When used downstream of a [type of current-limiting device] having [percent] % short-circuit impedance, this UPS is rated for use on a circuit capable of delivering no more than [short-circuit current] rms symmetrical amperes, [voltage] volts maximum."		N/A
	The first blank line ("type of current-limiting device") shall be filled in with either "reactor" or "transformer".		N/A
	The marked SHORT-CIRCUIT WITHSTAND RATING in the third blank line ("short-circuit current") shall be in accordance with the symmetrical values given in Table NNN.1 or Table NNN.2.		N/A
Annex OOO	reserved for future use		N/A
Annex PPP (normative)	Manufacturing and production tests		N/A
PPP.1	Production-Line Dielectric Voltage-Withstand Test		N/A
PPP.1.1	As a routine production-line test, each unit shall withstand without electrical breakdown the application of an alternating-current potential at a frequency within the range of 40 – 70 hertz or a direct-current potential:		N/A
	a) Between the primary wiring, including connected components, and accessible dead metal parts that are likely to become energized; and		N/A
	b) Between primary wiring and accessible low-voltage (42.4 volts peak, 60 volts, dc or less) metal parts, including terminals and connector contacts.		N/A
PPP.1.2	The test duration and potential shall be as described in either condition A or B of Table PPP.1.		N/A
PPP.1.3	The test potential can be gradually increased to the required value, but the full value is to be applied for 1 second or 1 minute, as required		N/A
PPP.1.4	The unit can be at intended operating temperature, at room temperature, or at any intermediate temperature for the test.		N/A
PPP.1.5	The test shall be conducted when the unit is fully assembled. It is not intended that the UPS be unwired, modified, or disassembled for the test except in the following cases.		N/A
	a) Parts such as snap covers or friction-fit knobs that interfere with performance of the test need not be in place.		N/A
	b) The test can be performed before final assembly if the test is equivalently representative of the completed unit. Any component not included shall		N/A

UL 1778			
Clause	Requirement + Test	Result - Remark	Verdict
	not affect the results with respect to determination of possible electric shock from miswiring, defective components, unacceptable spacings, and the like.		
	c)Solid-state components that might be damaged by a secondary effect (induced voltage surge, excessive heating, and the like) of the test may be short-circuited by means of a temporary electrical jumper, or the test may be conducted without the component electrically connected, provided that the wiring and terminal spacings are maintained. Additionally, surge protective devices other than capacitors connected from primary wiring to dead metal may be disconnected during the test.		N/A
PPP.1.6	The test equipment shall have a means of indicating the test potential, an audible or visual indicator of electrical breakdown and, for automated or station type operations, either a manual-reset device to restore the equipment after electrical breakdown or an automatic-reject feature for any		N/A
	unacceptable unit. When an alternating-current test potential is applied, the test equipment shall include a transformer having an essentially sinusoidal output.		N/A
PPP.1.7	When the rated output of the test equipment is less than 500 volt-amperes, the equipment shall include a voltmeter in the output circuit to indicate directly the applied test potential.		N/A
PPP.1.8	When the rated output of the test equipment is 500 volt-amperes or more, the test potential can be indicated:		N/A
	a)By a voltmeter in the primary circuit or in a tertiary-winding circuit;		N/A
	b)By a selector switch marked to indicate the test potential; or		N/A
	c)By a marking in a readily visible location to indicate the test potential in the case of equipment having a single test-potential output.		N/A
	If an indicating voltmeter is not used, the test equipment shall include a visual means (such as an indicator lamp) to indicate that the test voltage is present at the test-equipment output.		N/A
PPP.1.9	Test equipment other than that described in PPP.1.6 – PPP.1.8 may be used if found to accomplish the intended factory control.		N/A
PPP.1.10	For the test, either a sufficient number of control devices are to be closed or separate applications of the test potential are to be made so that all parts of the primary circuit are tested.		N/A
PPP.2	Production-Line Grounding-Continuity Test		N/A
PPP.2.1	Each unit that has a power-supply cord with a grounding conductor and each unit shipped without a power supply cord shall be tested, as a routine production-line test, to determine that grounding continuity is provided between the grounding blade or pin of the attachment plug and the accessible dead metal parts of the unit that are likely to		N/A

UL 1778			
Clause	Requirement + Test	Result - Remark	Verdict
	become energized. For a unit shipped without a power supply cord, the test is to be made between the grounding terminal of the cord connector and accessible dead metal parts.		
PPP.2.2	Only a single test need be made if the accessible metal selected is conductively connected to all other accessible metal.		N/A
PPP.2.3	Any indicating device (an ohmmeter, a battery and buzzer combination, or the like) can be used to determine compliance with the grounding continuity requirement.		N/A

UL 1778			
Clause	Requirement + Test	Result - Remark	Verdict

Critical Components

TABLE: List of critical components					P
Object/part No.	Manufacturer/ trademark	Type/model	Technical data	Standard (Edition / year)	Mark(s) of conformity ¹⁾
Fuse(F1, F2, F3)	Advanced Surge Tech Materials Ltd.	PB01040- 40A-2410	40A, 72V	UL	UR/UL Certificate no: E355868 (Link)
MCU(U4)	Shenzhen Branch of Shanghai Fudan Microelectronics Group Co., Ltd.	FM33LC026 N	FM33LC0xxx series low-power MCU, based on ARM Cortex-M0 kernel, integrated with large capacity embedded flash memory, with rich mode Analog and digital peripheral, and has excellent low power characteristics. □ Operating voltage range: 1.8V~3.6V (USB); 1.8V~5.5V (no USB) □ Operating temperature range: - 40℃~+85℃	UL	UR/UL
AFE(U1)	Sino Wealth Electronic Ltd.	SH367309	Hardware protection function Built-in balance switch forbid low voltage cell charging function Small current detection function supports disordered up and down power built-in watchdog module mode design-acquisition mode (SH367309 with MCU application) -protection mode (SH367309 independent application) - warehouse operation mode- burning mode	UL	UR/UL
LDO(U2)	Huada SEMICONDUCT OR Co., Ltd.	BL24C512A	Operating voltage 1.7V- 5.5V • Standard IIC interface communication protocol • Hardware write protection • Internal power-on reset • Maximum 1M working frequency • Ultra-low power design, static power consumption less than 1uA • Additional ID memory, providing chip ID and product traceability • 4 million erasure times; Data retention time more than 100 years •HBM ESD 8KV	EN 62040	Tested with a pliance

UL 1778			
Clause	Requirement + Test	Result - Remark	Verdict

NTC(RT1, RT3, RT4)	Shenzhen Topos Sensor Technology Co., Ltd.	TPS1-343F103F-100	temperature range: -40℃~+85℃-30~105	EN 62040	Tested with a ppliance
High speed dual channel digital isolator(U6, U11)	Shanghai Chuantu Microelectronics Co., Ltd.	CA-IS3721	• Signal transmission rate: DC to 150Mbps • Wide supply voltage range: 2.5V to 5.5V • Wide temperature range: -40° C to 125° C • Initialization is not required • Default output high level and low level options • Excellent electromagnetic immunity • Isolation voltage up to 5KVRMS • Isolation grid life: >40 years • Schmidt trigger input • Narrow-body SOIC8(S), wide-body SOIC8(G) package and wide-body SOIC16(W), in line with RoHS standards	EN 62040	Tested with a ppliance
RS-485 Transceiver(U8)	3peak Incorporated	TP8485E	• Exceeds Requirements of EIA-485 Standard • Hot Plug Circuitry-Tx and Rx Outputs Remain Three-State During Power-up/Power-down • Data Rate: Up to 250 kbps • Full Fail-safe (Open, Short, Terminated)Receivers • Up to 256 Nodes on a Bus (1/8 unit load) • Wide Supply Voltage 3V to 5.5V • SOIC-8 Package for Backward Compatibility • Bus-Pin Protection: ± 12 kV HBM protection	EN 62040	Tested with a ppliance

UL 1778			
Clause	Requirement + Test	Result - Remark	Verdict

N-MOSFET (Q2-11)	Shenzhen Sky Semiconductor Co., Ltd.	SS027N08LS	1、Package TOLL-8 2、Configuration Single 3、MOSFET Type N 4、VDS(V) 85、VGS(V) ± 20 、 ID(A) 240、 RDS(ON)(m Ω)(Typ./Max.)10V : 2.2/2.7	EN 62040	Tested with a ppliance
PCB (BMS)	HUIZHOU TAISHENG ELECTRONICS CO LTD	EMB08S008	Min. V-0, Min. 130° C	UL	E361828
PCB	MEIZHOU TONGZHENG ELECTRONIC CO LTD	EC-1	Min. V-0, Min. 130° C	UL94	UL E495295
Bridge Rectifier BR1	YANGZHOU KANGHONG SEMICONDUCTOR CO.,LTD	GBU10JE	Min.10A, min. 600V IEC 62368-1	EN 62040	Tested with a ppliance
Bridge Rectifier DB1	Shenzhen Jixin Micro Technology Co., Ltd.	DB207S	Min.2A, min.1000V IEC 62368-1	EN 62040	Tested with a ppliance
Bulk capacitor C30	Dongguan Chuanghui Electronic Factory	314V 470UF	470 μ F, Min. 3150V Min. 105° C IEC 62368-1	EN 62040	Tested with a ppliance
Bulk capacitor (Alternative) C30	Dongguan Guike Electronics Co., Ltd.	314V 470UF	470 μ F, Min. 3150V Min. 105° C IEC 62368-1	EN 62040	Tested with a ppliance
RELAY K2	HASCO RELAYS AND ELECTRONICS INTERNATIONAL CORP	SPRF1C16 DC12K	20A/277VAC 16A/250VAC 16A/24VDC Class B:-40° C~85° C	UL	UL E75887 TüV:R 50215857
Bulk capacitor C96	Dongguan Chuanghui Electronic Factory	200V 22UF	22 μ F, Min. 200V Min. 105° C IEC 62368-1	EN 62040	Tested with a ppliance
Bulk capacitor C37, C126, C127	Dongguan Chuanghui Electronic Factory	450V 6u8	6.8 μ F, Min. 450V Min. 105° C IEC 62368-1	EN 62040	Tested with a ppliance
Y capacitor C104	Welson Industrial Co., Ltd.	KL	2200pF, 250Vac, Y2 type, 125° C	DIN EN 60384-14	UL E104572 VDE 40016156

UL 1778			
Clause	Requirement + Test	Result - Remark	Verdict

Y capacitor C83, C88, C92, C125		Welson Industrial Co., Ltd.	KL	4700pF, 300Vac, Y2 type, 125° C	DIN EN 60384- 14	UL E104572 VDE 40016156
Y Capacitor C28		JYH CHUNG ELECTRONICS CO LTD	JY	6800PF, 300Vac, Y2 type, 125° C	DIN EN 60384- 14	UL E187963 VDE 123326
X Capacito C79		DONG GUAN HONGFARAD ELECTRONICS CO LTD	HMKP	0.0047uF, X2 type, Min. 310Vac, 110° C	DIN EN 60384- 14	UL E484578 VDE 40044173
X Capacito C99		DONG GUAN HONGFARAD ELECTRONICS CO LTD	HMKP	0.001uF, X2 type, Min. 310Vac, 110°C	DIN EN 60384- 14	UL E484578 VDE 40044173
X Capacito C106		DONG GUAN HONGFARAD ELECTRONICS CO LTD	HMKP	3.3uF, X2 type, Min. 310Vac, 110° C	DIN EN 60384- 14	UL E484578 VDE 40044173
CBB Capacitor C73		DONG GUAN HONGFARAD ELECTRONICS CO LTD	HMKP	10uF, Min. 600V Min. 105° C IEC 62368-1	EN 62040	Tested with a ppliance
Inducto r	L3	DONGGUANJUN JIAELECTRONIC STECHN OLOGY CO LTD	SBI4.2	CLASS 130(B) SBI4.2	UL1446	UL E348566
	Magnet wire	HENG YA ELECTRIC (DONGGUAN) LTD	*EIW/180, QZY-*/180	Min.180°C	UL1446	UL E197768
	EPOXY	DONGGUAN EATTO ELECTRONIC MATERIAL CO LTD	3300A- 1/3300B-1 BK	Min. 130° C	UL94	UL E218090
	EPOXY PLATE	HON TAI MATERIAL CO LTD	HT-E3502 FR-4.0	Min. 130° C	UL94	UL E122230
	VARNI SH	Dongguan Dabang Electronic Technology Co Ltd	DB-802 MW28-C	Min.155°C	UL94	UL E511289
Inducto r (Altern ative)	L3	DONGGUANJUN JIAELECTRONIC STECHN OLOGY CO LTD	SBI4.2	CLASS 130(B) SBI4.2	UL1446	UL E348566
	Magnet wire	TAI-I ELECTRIC WIRE&CABLE CO LTD	PEW-H	Min.180°C	UL1446	UL E197768

UL 1778			
Clause	Requirement + Test	Result - Remark	Verdict

	TEFLO N TUBE	GREAT HLODING INDUSTRIAL CO.,LTD.	VW-1 TFL	Min.200℃	UL94	UL E156256
	EPOXY	DONGGUAN EATTO ELECTRONIC MATERIAL CO LTD	E-500 94V-0	Min.130℃	UL94	UL E218090
	EPOXY PLATE	KINGBOARD LAMINATES HOLDINGS LTD	FR-4 94V-0	Min.130℃	UL94	UL E123995
Inducto r	LF1,LF 3	Guangzhou Poojin Electronic Co., Ltd.	TD1515-2mH	CLASS 130(B)	EN 62040	Tested with a ppliance
	Magnet wire	Guangdong liziwire Electric Technology Co Ltd	*UEW/NY/180, QPN/180, TQPN/180	Min.180℃	UL1446	UL E504985
	BOBBI N	CHANG CHUN PLASTICS CO LTD	T375HF	Min. 150° C	UL94	UL E59481
	VARNI SH	QUALIPOLY CHEMICAL CORP	1032(@)	Min.180℃	UL94	UL E228349
Inducto r	LF2	Guangzhou Poojin Electronic Co., Ltd.	TD1010-10mH	CLASS 130(B)	EN 62040	Tested with a ppliance
	Magnet wire	Guangdong liziwire Electric Technology Co Ltd	*UEW/NY/180, QPN/180, TQPN/180	Min.180℃	UL1446	UL E504985
	BOBBI N	CHANG CHUN PLASTICS CO LTD	T375HF	Min. 150° C	UL94	UL E59481
	VARNI SH	QUALIPOLY CHEMICAL CORP	1032(@)	Min.180℃	UL94	UL E228349
	VARNI SH	ELANTASELECT RICALINSULATI ON ELANTAS PDG INC	V1380FC	Min.130℃	UL94	UL E75225
Transfo rmer	T1	DONGGUANJUN JIAELECTRONIC STECHN OLOGY CO LTD	SBI4.2	CLASS 130(B) SBI4.2	EN 62040	UL E348566
	Magnet wire	TAIYI COPPER GUANGZHOU CO., LTD	UEW-F	Min.155℃	UL1446	UL E234896

UL 1778			
Clause	Requirement + Test	Result - Remark	Verdict

	BOBBIN	CHANG CHUN PLASTICS CO LTD OR EQU	T375HF 94V-0	Min.150℃	UL94	UL E59481
	MYLAR TAPE	JINGJIANG YAHUA PRESSURE SENSITIVE GLUE CO LTD OR EQU	CT-280	Min.130℃	UL1446	UL E165111
	MARGIN TAPE	JINGJIANG YAHUA PRESSURE SENSITIVE GLUE CO LTD OR EQU	WF* (c)(h)	Min.130℃	UL1446	UL E165111
	EPOXY	DONGGUAN EATTO ELECTRONIC MATERIAL CO LTD	E-500 94V-0	Min.130℃	UL94	UL E218090
	TEFLON TUBE	GREAT HLODING INDUSTRIAL CO.,LTD.	VW-1 TFL	Min.200℃	UL94	UL E156256
	VARNISH	ELANTASELECT RICALINSULATION ELANTAS PDG INC	V1380FC	Min.130℃	UL94	UL E75225
Transformer	T2	DONGGUANJUN JIAELECTRONIC STECHNOLOGY CO LTD	SBI4.2	CLASS 130(B) SBI4.2	UL	UL E348566
	BOBBIN	CHANG CHUN PLASTICS CO LTD OR EQU	T375HF 94V-0	Min.150℃	UL94	UL E59481
	Magnet wire	TAIYI COPPER GUANGZHOU CO., LTD	UEW-F	Min.155℃	UL1446	UL E234896
	MYLAR TAPE	JINGJIANG YAHUA PRESSURE SENSITIVE GLUE CO LTD OR EQU	CT-280	Min.130℃	UL94	UL E165111
	MARGIN TAPE	JINGJIANG YAHUA PRESSURE SENSITIVE GLUE CO LTD OR EQU	WF* (c)(h)	Min.130℃	UL94	UL E165111

UL 1778			
Clause	Requirement + Test		Verdict

	VARNISH	ELANTASELECT RICALINSULATION ELANTAS PDG INC	V1380FC	Min.130°C	UL94	UL E75225
	EPOXY	DONGGUAN EATTO ELECTRONIC MATERIAL CO LTD	E-500 94V-0	Min.130°C	UL94	UL E218090
FUSE F3,F4		DONG GUAN ANDU ELECTRONICS CO LTD	3GTU/3GTV	5A, 250V	UL248	UL E75225
F1,F2		LITTEL FUSE INC	MIN	30A 32V	UL248	UL E71611
F1,F2 (Alternative)		DONG GUAN ANDU ELECTRONICS CO LTD	ADATC-030	30A 32V	UL248	UL E317400
Optocoupler U1, U3, U5, U20		Everlight Electronics Co., Ltd.	EL817	Di ≥0.4mm, Ex ≥7.0mm, 110° C	DIN EN IEC 60747-5-5 UL1577	UL E214129 VDE 132249
Optocoupler U12,U14, U15,U16		Suzhou Novosense Microelectronics Co.,Ltd	NSI6801TC-DDBR	Di ≥0.4mm, Ex ≥7.0mm, 150°C	UL1577	UL E500602
VARISTOR MOV1		SHENZHEN CHUANG NA ELECTRONICS CO LTD	CVR-14D471K CVR-10D471KT	470V, 105° C	DIN EN 61051-1	UL E477910 VDE 40043102
Thermistor NTC1		NANJING SHIHENG ELECTRONICS CO LTD	MF72-10D9	10R, 2A	EN/IEC: IEC 60539-1	UL E241319 TUV-RH: R 50245892
Thermistor NTC1 (Alternative)		Nanjing KeMin Electronics Co., Ltd.	MF72-10D9	10R, 2A	EN/IEC: IEC 60539-1	TUV-SUD: B151093844
Lead wire X1,X2		GuangDongTeCheng Metal & Electronics Co Ltd	1015	600Vac, 105° C, VW-1, Min. 10AWG	DIN EN 60384-14	UL E249743
NTC	NTC	AnHui Eagle Electron Co.,Ltd	MF58-503J 3950YA	Class C2 Tmoa (° C) 200° C	UL 1434	UL E502983

UL 1778			
Clause	Requirement + Test	Result - Remark	Verdict

	NTC lead wire	DONGGUAN WORLDFUL ELECTRONIC WIRE CO.,LTD	UL2651	Min .26AWG Min.300V Min.105° C PVC-VW-1 UL758	UL758	UL E317806
	EPOXY	DONGGUAN EATTO ELECTRONIC MATERIAL CO LTD	3300A-1/3300B-1 BK,YL	Min. V-0, Min. 130° C	UL94	UL E218090
CELL						
	Wiring for P+, P-	DONGGUAN ZHONGZHEN ENERGY TECHNOLOGY CO.,LTD	1015	10AWG, 200°C, 600V	UL 758	UL E355578
	Wiring for P+, P- Alternative)	Interchangeable	1015	10AWG, 200°C, 600V	UL 758	UL Approved
	Wiring for L+, L-	ShenZhen Greent on Electronic Co. , Ltd.	1322	16AWG, 200°C,300V	UL 758	UL E465814
	Wiring for L+, L- Alternative)	Interchangeable	1322	16AWG, 200°C,300V	UL 758	UL E465814
Supplementary information: ¹⁾ Provided evidence ensures the agreed level of compliance.						

UL 1778			
Clause	Requirement + Test	Result - Remark	Verdict

TABLE: electric shock and energy hazards						P
Supply Voltage	Location (e.g. circuit designation)	Test conditions	Parameters			
			U (V)	I (mA)	Additional Info 2)	
140Vac, 60Hz	Secondary data output port	Normal	224Vdc	-	-	
		U1 Short circuit	213Vdc	-	-	
		Q1 Short circuit	215Vdc	-	-	
Supplementary information:						

TABLE: Provisions for earthing and bonding					P
Location	Test current (A)	Duration (min)	Voltage drop (V)	Resistance (Ω)	
GND pin of AC Inlet and the farthest Plastic enclosure	40	2	1.5	0.032	
Supplementary information: --					

	TABLE: Overcurrent and earth fault protection in primary circuit		P
Supply voltage (V).....:	140		—
Phase(s)	[X] Single Phase; [] Three Phase: [] Delta [] Wye		—
Power Distribution System	☒ TN ☐ TT ☐ IT		—
Location	Touch current (mA)	Comment	
L/N to Plastic enclosure	0.05		
L/N to output terminal	0.09		
Supplementary Information: --			

Thermal requirements				☐ N/A			☐ Fail		☒ Pass	
VOLTAGE (V)		140		INPUT (W)		938				
CURRENT (A)		4.61		FREQ. (Hz)		50Hz				
AMB. TEMP.		t1 = 24,3 °C (before test)		t2 = 24,2 °C (after test)						
Ch. No.	Location / Part (by thermocouple)			Rise (K)		Limit (K /°C)		Pass	Fail	
	PCB			66.7		130		✓		
	Internal wire			39.3		80		✓		

UL 1778					
Clause	Requirement + Test	Result - Remark			Verdict
	X capacitor	31.2	125	✓	
	Transformer winding	74.4	110	✓	
	Transformer bobbin	72.6	Ref	✓	
	Plastic enclosure	35.6	65	✓	

Test Between	Test Voltage (V)	Result		
		P	F	NA
Live parts and accessible dead metal parts	1528	☒	☐	☐

-Appendix 1: Photo document



Photo 1: Overall view



Photo 2: Internal view

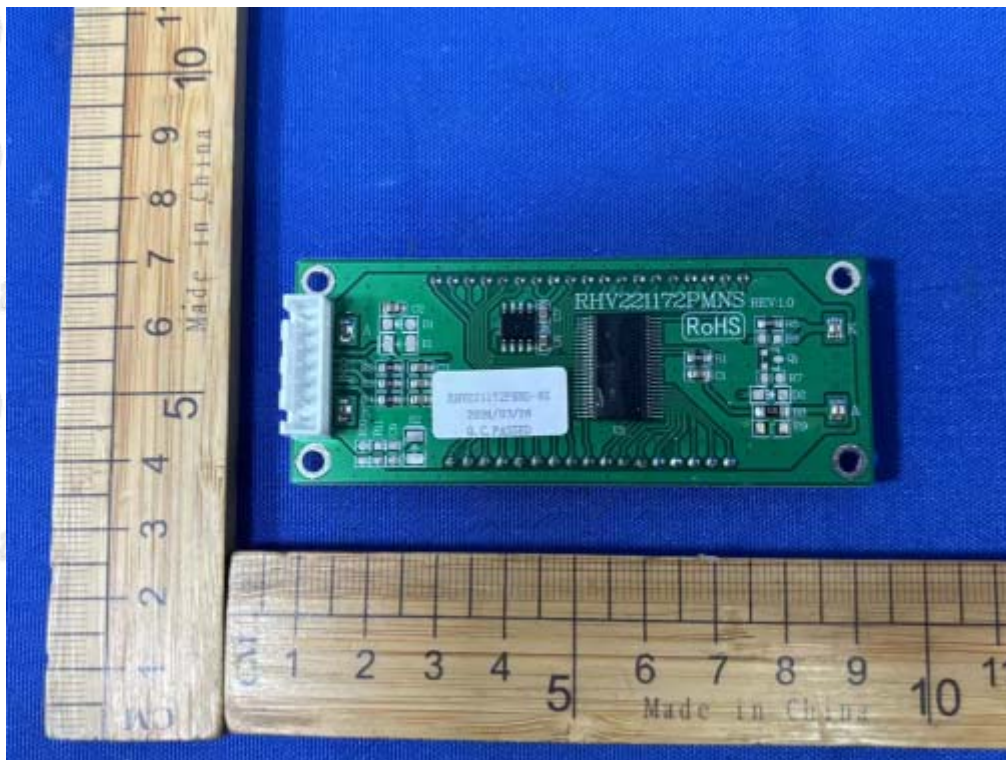


Photo 3: PCB view

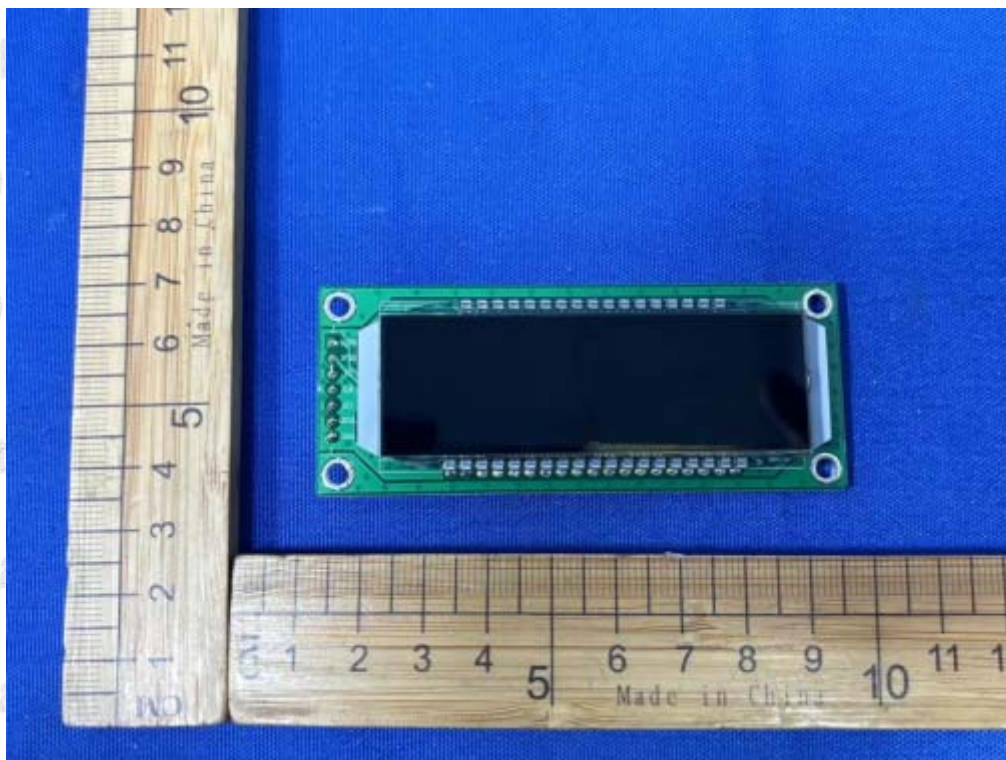


Photo 4: PCB view

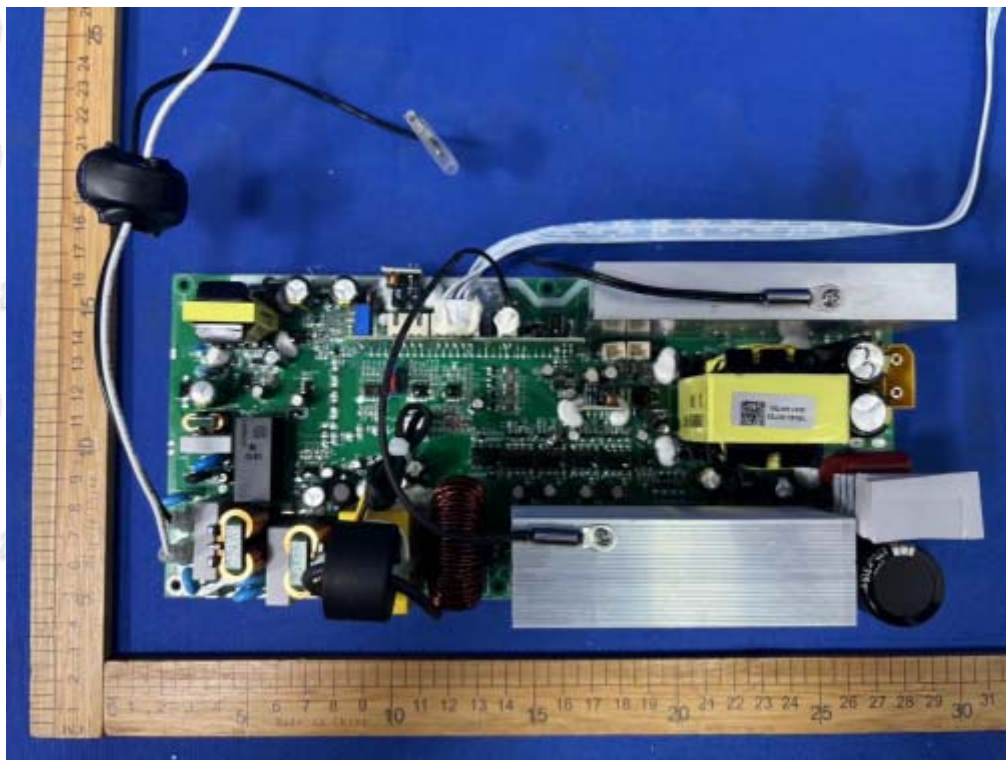


Photo5: PCB view

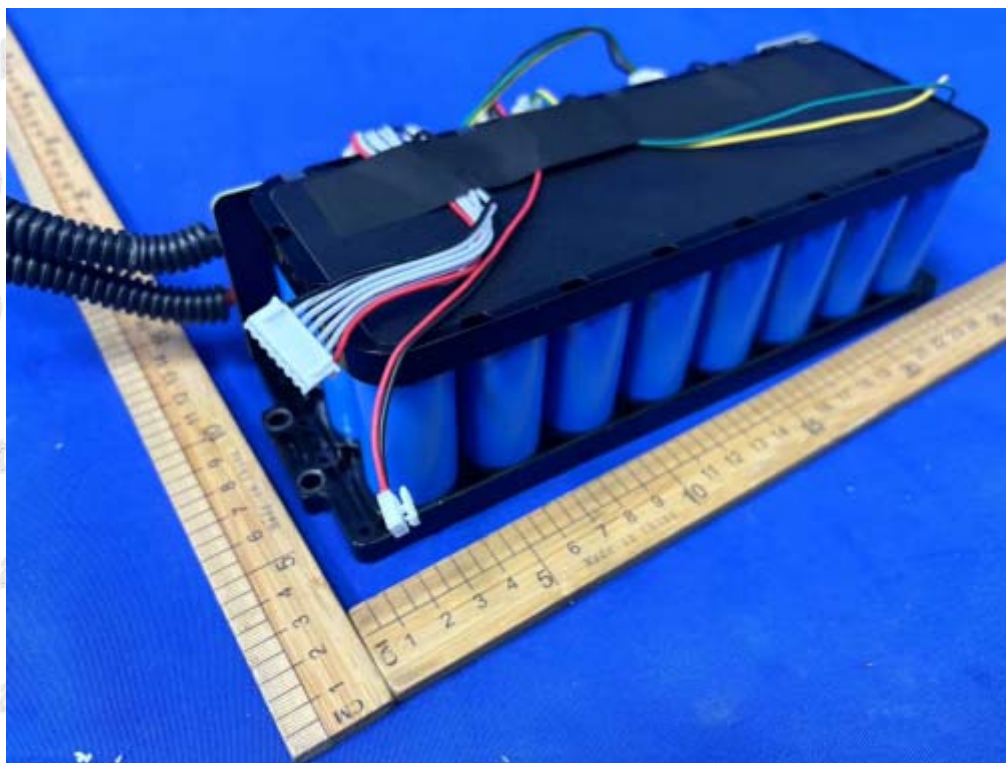


Photo6: Cell view

----- End of report -----